# W-SpectralFormer: A Physics-Informed Transformer Model for Weakly Supervised Spectral Super-Resolution of High-Fidelity Remote Sensing Reflectance Over Inland Lakes

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***Abstract*—The inherent tradeoff between spatial and spectral resolution in remote sensing sensors limits their ability to monitor inland lake eutrophication. Spectral super-resolution (SSR) provides a cost-effective solution for obtaining remote sensing images with high spatial and spectral resolution. However, the water-leaving remote sensing reflectance is characterized by weak signals and rich spectral features related to absorbing and scattering effects of water components. Without full consideration of data characteristics and proper physical modeling of the SSR process, existing methods may introduce significant uncertainties in reconstructed remote sensing reflectance ($R_{rs}$) over inland waters. To this end, we propose a novel physics-informed SSR framework (referring to W-SpectralFormer) for Sentinel-2 multispectral instrument (MSI) images that achieves high-fidelity spectrum, where the core idea is to integrate optical properties of water to guide the feature learning process. With Sentinel-3 Ocean and Land Colour Instrument (OLCI) data serving as the spectral reference, W-SpectralFormer features a frequency-assisted transformer-based SSR (FATSSR) network that is optimized through weak supervision using a pretrained degradation estimator. The FATSSR network employs an F-token module that embeds the fast Fourier transform (FFT) into the Transformer tokens, enabling localized attention to reconstruct vital spectral features. Moreover, a spectral-aware (SA) loss function is devised to incorporate both inherent and apparent optical properties of water as constraints during model training for enhanced spectral refinement. Both the F-token module and SA loss coupled to boost the accuracy of the reconstructed spectrum in inland waters with complex optical properties. Experiments demonstrate that the proposed method achieves state-of-the-art SSR performance. Moreover, validation of chlorophyll-a (Chla) concentrations confirms that the reconstructed images provide more reliable results than the original MSI data with a coarser spectral resolution.**

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***Index Terms*—Fast Fourier transform (FFT), optical properties, Sentinel-2 multispectral instrument (MSI), Sentinel-3 Ocean and Land Colour Instrument (OLCI), spectral super-resolution (SSR).**

## I. Introduction

INLAND rivers and lakes are essential natural resources for the survival and development of human society [1]. Amidst rapid social development and global climate change, inland waters are experiencing significant deterioration [2]. As a result, water quality degradation has raised global concerns, particularly for inland lakes and reservoirs in densely populated regions. In addition to the traditional in situ monitoring solutions, remote sensing reflectance ($R_{rs}$) records the optical properties of water constituents and is capable of providing large-scale and periodical observations of optically active water quality indicators (WQIs). Therefore, extensive studies have been conducted to use remote sensing data for monitoring these WQIs in inland waters on both local and global scales [3], [4], [5].

As a primary indicator of phytoplankton biomass, chlorophyll-a (Chla) concentration provides a reliable measure of water body eutrophication status [6], [7]. The retrieval of Chla concentration using remote sensing data primarily depends on the spectral absorption valley near 665 nm and the scattering characteristics near 709 nm. In turbid Case II waters, narrow spectral bands are particularly crucial for Chla estimation to reduce the influence of complex constituents in waters (e.g., suspended particulate matter (SPM) and colored dissolved organic matter (CDOM)) on the featured signals in the spectra.

Satellite remote sensing is a key technology for monitoring Chla concentrations in inland waters, but there exists an inherent tradeoff in the performance of existing sensors. Ocean color sensors, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) [8] and the Ocean and Land Colour Instrument (OLCI) [9], offer suitable spectral configurations but have coarse spatial resolutions ($\geq$ 250 m), making it challenging to monitor medium- and small-sized water bodies. On the other hand, terrestrial sensors, such as the operational

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land imager (OLI) onboard the Landsat 8/9 satellites and the multispectral instrument (MSI) onboard the Sentinel-2 satellites, provide higher spatial resolutions ≤ 30 [10]. However, their spectral design is not optimized for the complex optical characteristics of turbid waters, which may bring a challenge in Chla retrieval in inland water environments [11]. To address the tradeoff between spatial and spectral resolution in satellite sensors, the spectral super-resolution (SSR) technique provides a cost-effective solution. Specifically, SSR generates images with high spatial and spectral resolution by reconstructing spectral bands while preserving spatial information [12], [13], [14]. Compared to spatio-spectral fusion methods that require image pairs with complementary spatial and spectral information as the model input [15], SSR receives a single image with coarse spectral bands as the input and generates the result imagery with an enhanced spectral resolution. The essence of SSR is an ill-posed problem, which requires prior knowledge as a constraint of the model solution. Early SSR methods relied on linear interpolation or sparse coding to model spectral mapping relationships between input and output images [16]. However, these approaches were limited to linear approximations of coarse-to-fine spectral transformations, often failing to accurately reconstruct complex spectral features across diverse ground objects.

In recent decades, deep learning-based methods have made remarkable progress for the task of SSR due to their excellent representation and fitting abilities. The goal of deep learning-based SSR methods is to learn the optimal representation between images with coarse and fine spectral resolutions for the recovery of spectral bands. Among these approaches, models based on convolutional neural networks (CNNs) are widely adopted and have achieved promising performance [17], [18]. Similarly, transformer-based models have demonstrated excellent results and gained significant popularity [19], [20], [21]. More recently, researchers attempted to incorporate imaging physics in the feature learning process for remote sensing spectral reconstruction. For example, Li et al. [22] proposed a new adaptive weighted attention network (AWAN) that reallocates channel feature weights by exploiting the correlation between channels to achieve accurate spectral reconstruction. Cai et al. [23] proposed a spectral-wise multihead self-attention (S-MSA) model, which treats each spectral channel as a token and calculates self-attention along the spectral dimension. Wu et al. [24] introduced a semantic-driven spatial relation module (SSRM) that leverages RGB semantic information to adaptively aggregate features from semantically consistent regions to enhance spectral reconstruction accuracy.

Although numerous deep learning methods have been proposed for the SSR of remote sensing images (RSI), most of them are trained in a supervised manner, which requires images with both high spatial and spectral resolution as the reference. However, acquiring high-resolution reference images with fine spectral bands for model training is challenging in practice, limiting the applicability of fully supervised methods.

Furthermore, the light propagation process in underwater environments is highly complex, with water surface reflectance accounting for only about 10% of the total reflected signal received by imaging sensors [25]. Therefore, the direct application of general SSR methods for water body reconstruction is limited. $R_{rs}$ is highly susceptible to interference from uncertain factors during the spectral reconstruction process. This can lead to distortion of the spectral features that reflect the complex optical characteristics of water body components, thus reducing the accuracy of derived WQIs [26]. For inland water monitoring applications, the SSR method needs to carefully integrate complementary spatial–spectral information from available data sources during the modeling process, while fully considering the physical principles of the radiative transfer process in water bodies to achieve high-fidelity spectral reconstruction.

To address the above issues, we propose a physics-informed weakly supervised SSR framework for spectral reconstruction of $R_{rs}$ over inland waters, referring to W-SpectralFormer. The core idea of W-SpectralFormer is to incorporate the optical properties of water as constraints within the Transformer-based network for high-fidelity spectral reconstruction. Specifically, the W-SpectralFormer employs the OLCI data as the spectral reference, and the spatial and spectral mapping relations between MSI and OLCI data are constructed via a pretrained degradation estimator. In this way, the weakly supervised SSR is achieved without real labels with both high spatial and spectral resolution. To enhance the reconstruction of spectral features, the fast Fourier transform (FFT) is embedded into the self-attention tokens, enabling the frequency-assisted transformer-based SSR (FATSSR) network to prioritize high-frequency spectral variations in attention-weighted feature learning. Moreover, a novel spectral-aware (SA) loss function is devised to use the optical properties of water as physical constraints to the reconstructed spectrum. Our main contributions can be summarized as follows.

1) We propose a weakly supervised SSR framework that reconstructs high-fidelity $R_{rs}$ over inland waters by leveraging OLCI spectral references and establishing MSI-OLCI spatial mapping via a pretrained degradation estimator.
2) We propose FATSSR, a physics-informed transformer-based network for SSR that incorporates FFT into the self-attention tokens to selectively enhance spectral details during reconstruction.
3) By integrating constraints on both inherent and apparent optical properties (IOPs and AOPs) of water, the SA loss function is devised to enable effective learning of complex spectral characteristics of inland waters for precise spectral refinement.

## II. Materials and Methods

### A. Study Area

This study focuses on datasets covering three typical eutrophic lakes located in the middle and lower reaches of the Yangtze River, which include Lake Donghu, Lake Liangzihu, and Lake Chaohu (Fig. 1). These lakes perform critical ecological and socioeconomic functions, including water supply, flood regulation, aquaculture, and biodiversity conservation.

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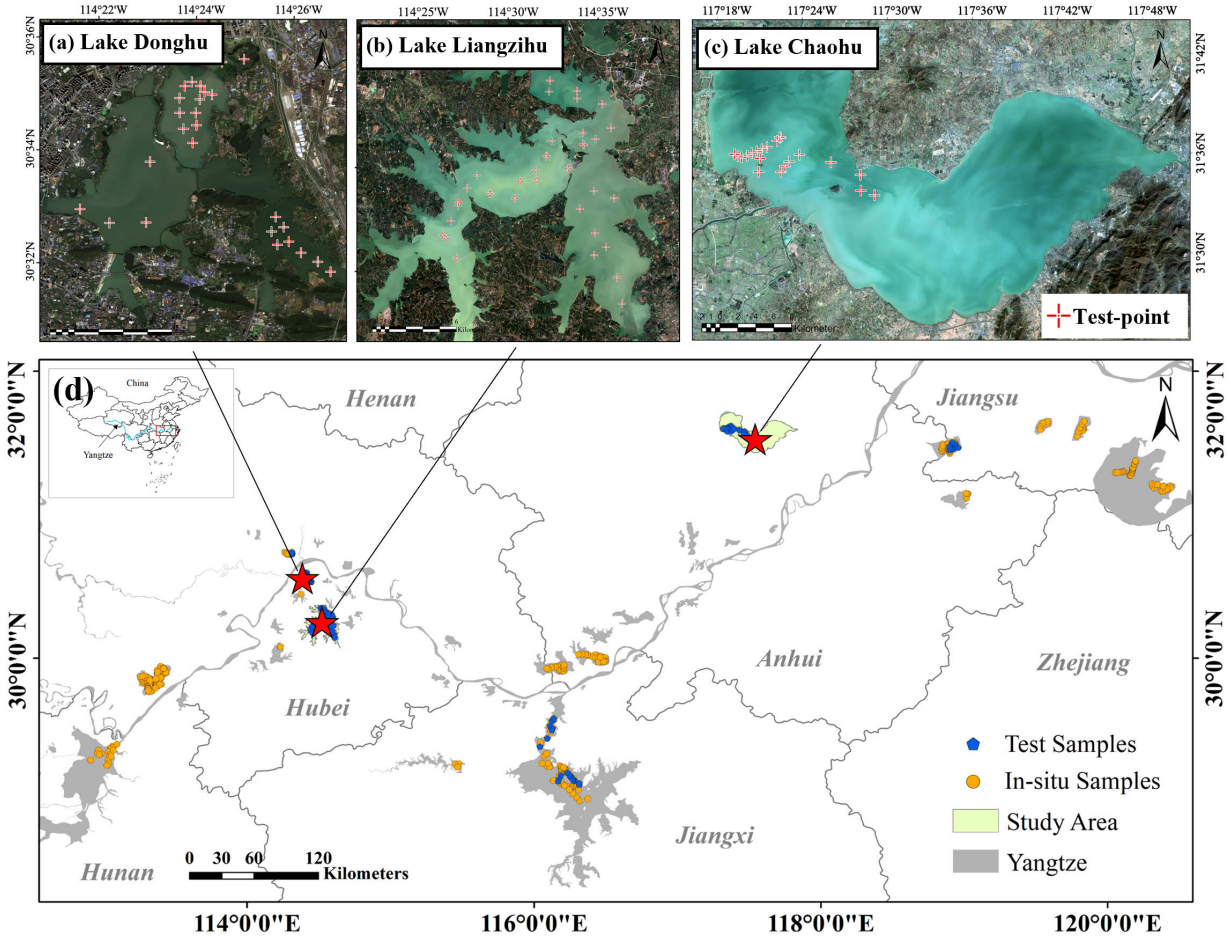



Fig. 1. Information of the study area used in this study. (a)–(c) True-color images of Lake Donghu, Lake Liangzihu, and Lake Chaohu, respectively. (d) Location of the study sites, along with the spatial distribution of 426 hyperspectral $R_{rs}$ and corresponding Chla measurements as the training and test dataset for the inversion model.

TABLE I

GEOGRAPHIC LOCATIONS AND PERMANENT-WATER COVERAGE AREAS OF THE STUDY LAKES. THE LAKE AREAS ARE DERIVED FROM THE SURFACE WATER OCCURRENCE (SWO) DATA WITHIN 1984 TO 2019 IN THE GLOBAL SURFACE WATER DATASET (GSWD) [28], WHICH INDICATE THE PERMANENT WATER AREA WITH SWO >80%

| Region | Location | | Area($km^2$) |
|---|---|---|---|
| | Lat.(N) | Lon.(E) | |
| Lake Donghu | 30°30'–30°37' | 114°21'–114°28' | 23.63 |
| Lake Liangzihu | 30°04'–30°20' | 114°31'–114°42' | 221.86 |
| Lake Chaohu | 31°25'–31°43' | 117°16'–117°51' | 760.63 |

However, their aquatic ecosystems have faced increasing pressures in recent decades, primarily due to cumulative human activities and improper development and utilization [27]. Therefore, accurate and efficient monitoring of water quality is urgently needed to protect water environments and achieve sustainable water resource management. Detailed information of the study lakes is provided in Table I.

### B. In Situ Data

During 2018 to 2021, field surveys were carried out in 18 lakes across the middle and lower reaches of the Yangtze River, collecting 426 hyperspectral $R_{rs}$ measurements and corresponding Chla concentrations. The $R_{rs}$ measurements at the water surface were collected following the ocean optical protocol recommended by NASA [29]. The field portable spectrometers (PSR+ 3500 or PSR-1100f) from Spectral Evolution Inc. (SEI) were used to measure the total radiance at the water surface ($L_t$), downward sky radiance ($L_{sky}$), and standard reference patch radiance ($L_{pla}$). The $R_{rs}$ was then derived as

$$R_{rs} = \frac{\rho_{pla}\left(L_t - \rho_f \times L_{sky}\right)}{\pi L_{pla}} - \Delta \tag{1}$$

where $\rho_f$ is the Fresnel reflectance coefficient of the water surface assumed to be 0.022–0.028, as suggested in [29] and [30], and $\rho_{pla}$ is the standard reflectance of the reference panel. In (1), $\Delta$ represents the residual reflected skylight, which is calculated based on the relative height of the water-absorption-dip-induced reflectance-peak (RHW) at 810 nm [31]. For each sampling point, $R_{rs}$ ranges from 400 to 900 nm with a 1-nm interval. Moreover, the Chla concentration was analyzed by a spectrophotometer with the collected water samples in the laboratory [32]. The $R_{rs}$ measurements were simulated to generate simulated satellite bands for model validation, and the Chla dataset was utilized for training and testing the Chla inversion model (see Section IV-A2).

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TABLE II
CORRESPONDING BANDS OF SENTINEL-2A MSI AND SENTINEL-3A OLCI

| Band | Sentinel-2 MSI | | Sentinel-3 OLCI | |
|---|---|---|---|---|
| | Wavelength (nm) | Spatial resolution (m) | Wavelength (nm) | Spatial resolution (m) |
| Coastal | B1: 433–453 | 60 | Oa03: 437.5–447.5 | 300 |
| Blue | B2: 470–510 | 10 | Oa04: 485–495<br>Oa05: 505–515 | 300<br>300 |
| Green | B3: 543–578 | 10 | Oa06: 560–570<br>Oa08: 660–670 | 300<br>300 |
| Red | B4: 650–680 | 10 | Oa09: 670–677.5<br>Oa10: 677.5–685 | 300<br>300 |
| Red-edge 1 | B5: 698–713 | 20 | Oa11: 700–710 | 300 |
| Red-edge 2 | B6: 730–749 | 20 | Oa12: 750–757.5 | 300 |
| Red-edge 3 | B7: 773–793 | 20 | Oa15: 766.25–768.75<br>Oa16: 771.25–786.25 | 300<br>300 |
| NIR | B8: 773–907<br>B8A: 855–875 | 20<br>20 | Oa17: 855–875<br>Oa18: 880–890 | 300<br>300 |

TABLE III
DETAILED INFORMATION ABOUT THE IMAGE DATASETS USED IN THIS STUDY. IMAGES ARE CLIPPED TO ADAPT TO THE SIZE OF EACH LAKE

| Region | Datasets | | Image size (clipped) | |
|---|---|---|---|---|
| | Training | Testing | MSI | OLCI |
| Lake Donghu | 19 | 3 | 555×555 | 37×37 |
| Lake Liangzihu | 11 | 3 | 1800×1800 | 120×120 |
| Lake Chaohu | 7 | 2 | 1800×2800 | 120×188 |

### C. Satellite Data

The MSI is a multispectral imager onboard the Sentinel-2A (2015-), Sentinel-2B (2017-), and Sentinel-2C (2024-) satellites. It measures the Earth's radiation in 13 spectral bands from visible to near infrared (VNIR) and short-wave infrared (SWIR) ranges, with different spatial resolutions (10–60 m) across different bands. Each satellite has a revisit frequency of ten days, while the combined revisit cycle of two satellites can achieve a five-day interval. In this study, the MSI bands are resampled to 20 m for the convenience of processing. The OLCI instrument, carried onboard the Sentinel-3A (2016-) and Sentinel-3B (2018-) satellites, measures Earth's radiation across 21 spectral bands from VNIR to SWIR (400–1040 nm) with a spatial resolution of 300 m. The combined revisit period of the two Sentinel-3 satellites is less than two days.

This study collected synchronized and cloud-free MSI and OLCI images from the three study areas during 2021 to 2023 to build image pairs for model training and testing. Considering the sensitive band requirements for Chla inversion and the spectral accuracy after atmosphere correction, 9 bands (B1-B8A) were selected for MSI data, and 14 bands (Oa3–Oa12 and Oa15–Oa18) were selected for OLCI data. Detailed band information is provided in Table II. The images were processed using Acolite DSF atmospheric correction [33], followed by georeferencing and coordinate system unification. To highlight the water reflectance, the images were cropped based on the extent of the waterbodies. As a result, a total of 45 MSI-OLCI image pairs were generated, with the number and size of training and test data for each lake detailed in Table III.

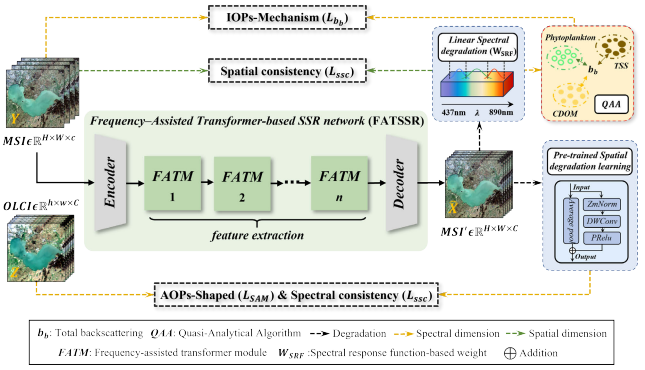



Fig. 2. Overall pipeline of the proposed W-SpectralFormer framework.

## III. METHODOLOGY

### A. Overall Framework of W-SpectralFormer

In this section, we present a weakly supervised SSR framework named W-SpectralFormer for addressing the challenge of reconstructing $R_{rs}$ over inland waters. As shown in 2, the goal of W-SpectralFormer model is to enhance the spectral resolution of MSI as $Y \in \mathbb{R}^{H\times W\times c}$ by leveraging the rich spectral information from OLCI as $Z \in \mathbb{R}^{h\times w\times C}$, and to reconstruct MSI′ as $\hat{X} \in \mathbb{R}^{H\times W\times C}$ with both high spatial resolution and narrow spectral bands. The spatial dimensions of the MSI and OLCI images are denoted as $H \times W$ and $h \times w$, while $c$ and $C$ represent the number of spectral bands in the MSI and OLCI images, respectively.

The W-SpectralFormer framework integrates three core components, which include: 1) the FATSSR network employing FFT-enhanced transformer self-attention for explicit high-frequency spectral reconstruction; 2) a pretrained degradation estimator that establishes MSI-OLCI spatial–spectral mapping constraints; and 3) the SA loss function leveraging aquatic optical properties to enhance model accuracy and generalizability. The simulation process of W-SpectralFormer can be expressed as

$$\hat{X} = \arg\min_{\hat{X}} \left\| Y - f_{D_{\text{spec}}}\left(\hat{X}\right) + \left(Z - f_{D_{\text{spa}}}\left(\hat{X}\right)\right) \right\|_{\text{SA-loss}} \quad (2)$$

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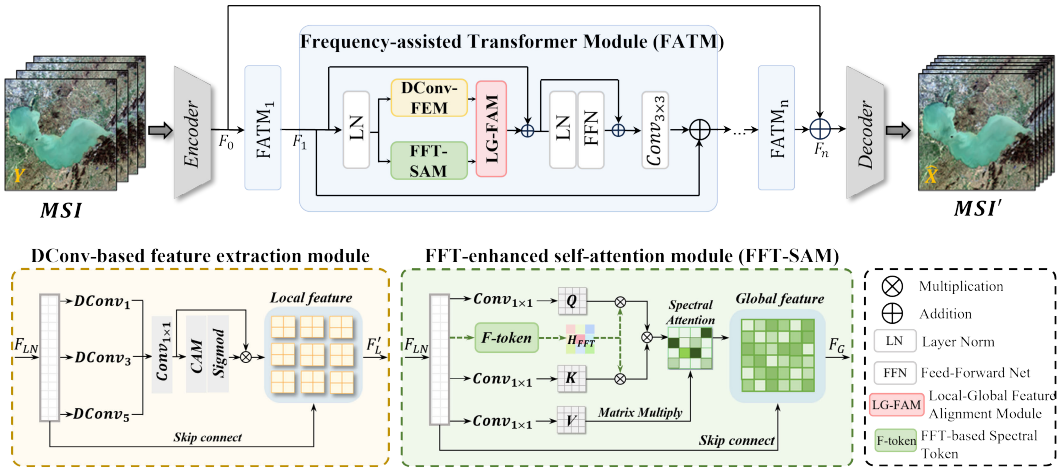



Fig. 3. Overall architecture of the proposed FATSSR.

where $f_{D_{\text{spec}}}(\cdot)$ denotes spectral degradation function, and $f_{D_{\text{spa}}}(\cdot)$ denotes spatial degradation function from the pretrained degradation estimator. A detailed explanation of each key component is provided below.

### B. FATSSR Network Architecture

As shown in Fig. 2, the FATSSR architecture consists of a simple backbone, which includes an encoder convolution block, multiple frequency-assisted transformer modules (FATMs), and a decoder convolution block. Specifically, the FATM implements multiscale feature extraction based on dilated convolution, coupled with an FFT-enhanced self-attention module (FFT-SAM) that amplifies high-frequency components within global feature representations. These spectrally enhanced features are then adaptively fused through a local–global feature alignment module (LG-FAM), thereby improving the reconstruction of fine-scale structures and spectrally rich details in water bodies. The structure of the FATM is displayed in Fig. 3.

The input MSI $Y \in \mathbb{R}^{H\times W\times c}$ is first processed with a convolutional feature encoding block composed of two $3\times 3$ convolutional layers and a parametric ReLU (PRelu) activation function. The FATMs are then applied to the encoded feature map $F_0$ for deep feature learning. This process can be expressed as

$$F_n = \text{FATM}_n\left(\text{FATM}_{n-1}\left(\cdots \text{FATM}_1\left(F_0\right)\right)\right) \tag{3}$$

where $n$ is the number of FATM, and $F_n$ is the output of the $n$th FATM. We incorporate a global skip connection to connect the initial feature $F_0$, enabling the network to preserve raw feature details while mitigating gradient vanishing in deep layers. At the tail of the FATSSR network, the decoder block with the same structure as the encoder block is employed to restore the spectral feature dimension of the super-resolved output

$$\hat{X} = \text{PReLU}\left(\text{Conv}_{3\times 3}\left(\text{Conv}_{3\times 3}\left(F_n + F_0\right)\right)\right) \tag{4}$$

where $\text{Conv}_{k\times k}(\cdot)$ represents the $k\times k$ convolutional layer. Notably, the following description details the feature processing in the first FATM ($\text{FATM}_1$), while all subsequent FATMs perform identical computational operations.

*1) Dilated Convolution-Based Feature Extraction Module:* To address the inherent limitations of transformer's global modeling in local feature extraction, we introduce the dilated convolution-based feature extraction module (DConv-FEM) to enhance spatial detail representation through hierarchical local feature extraction. As illustrated in Fig. 3, this module first normalizes the shallow feature $F_0$ through a layer normalization (LN) to obtain the intermediate feature $F_{\text{LN}}$. Three parallel DConv layers with progressive dilation rates (1, 3, and 5) then process $F_{\text{LN}}$ to capture multiscale spectral characteristics, generating hierarchical local features with enhanced spectral representation. Subsequently, a $1\times 1$ convolutional layer integrates and compresses these features while simultaneously performing concatenation and dimensionality reduction

$$\begin{aligned} F_L = \text{Conv}_{1\times 1}\left(\left[\text{DConv}_1\left(F_{\text{LN}}\right), \text{DConv}_3\left(F_{\text{LN}}\right)\right.\right. \\ \left.\left.\text{DConv}_5\left(F_{\text{LN}}\right)\right]\right) \end{aligned} \tag{5}$$

where $\text{DConv}_s(\cdot)$ represents the dilated convolution operation with the kernel size of $3\times 3$ and the dilation rate of $s$, $[\cdot,\cdot]$ denotes the concatenation operation, and $F_L$ encodes the dilated local spectral information.

Then, we employ a channel attention mechanism (CAM) [34] integrated with a Sigmoid function to capture cross-channel dependencies and weight their significance. This improves FATM's ability to focus on discriminative local spectral features

$$F'_L = F_L \otimes \text{Sigmoid}\left(\text{CAM}\left(F_L\right)\right) \tag{6}$$

where $F'_L$ represents the final dilated local spectral features extracted through the DConv-FEM, $\otimes$ indicates element-wise multiplication, and $\text{CAM}(\cdot)$ denotes the channel attention mapping operation.

*2) FFT-Enhanced Self-Attention Module:* The global channel reception of spectral-wise self-attention leads to spatially uniform spectral processing, potentially suppressing locally distinctive features that are critical for accurate water reflectance reconstruction. To address this, we introduce a

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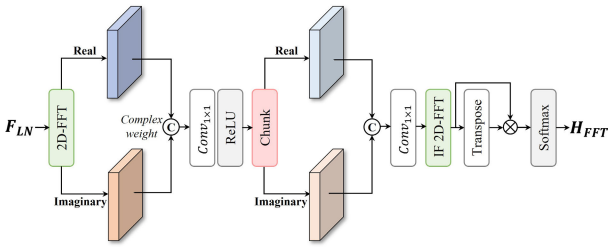



Fig. 4. Architecture of the F-token.

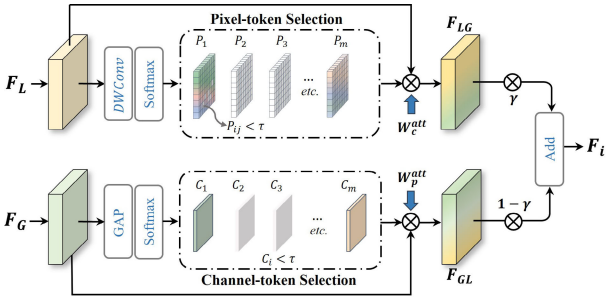



Fig. 5. Architecture of the LG-FAM.

novel spectral token derived from FFT decomposition, termed F-token. As illustrated in Fig. 4, the F-token incorporates convolutional operations in the frequency domain to separate high-frequency spectral variations, thereby enabling FFT-SAM to enhance spectral details while maintaining global coherence.

FFT is adopted due to its strong capability in capturing global spectral correlations and frequency-based structures [35]. In the F-token, the spectral features are first transformed into the frequency domain using a 2D-FFT, yielding complex-valued frequency components composed of real and imaginary parts. Learnable weights are independently applied to the real and imaginary components to adaptively model frequency bands that are highly correlated with water spectral details. Subsequently, the filtered frequency components are processed through lightweight $1 \times 1$ convolutional layers followed by ReLU nonlinear activations, thereby achieving adaptive selection and enhancement of frequency information based on the learned importance weights. The resulting high-frequency attention token $H_{\mathrm{FFT}}$ modulates both the query ($Q$) and key ($K$) matrices via element-wise multiplication, as illustrated in Fig. 3, thereby steering attention toward spectrally informative regions during reconstruction. The calculation can be expressed as follows:

$$\begin{aligned} Q &= \mathrm{Conv}_{1\times1}\left(F_{\mathrm{LN}}\right) \otimes H_{\mathrm{FFT}} \\ K &= \mathrm{Conv}_{1\times1}\left(F_{\mathrm{LN}}\right) \otimes H_{\mathrm{FFT}} \\ V &= \mathrm{Conv}_{1\times1}\left(F_{\mathrm{LN}}\right) \end{aligned} \tag{7}$$

$$F'_G = F_{\mathrm{LN}} \otimes \mathrm{Softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right) V \tag{8}$$

where $F'_G$ represents the global spectral information extracted through the FESM.

*3) Local–Global Feature Alignment Module:* In this section, we introduce the LG-FAM to automatically fuse global and local features, thereby improving contextual spectral information extraction (Fig. 5). In LG-FAM, the pixel-token selection computes the local weights $P_{\mathrm{ij}}$ using a $3 \times 3$ depth-wise convolution (DWConv) layer and a sigmoid function, then masks out the weights below the threshold $\tau = 1 \times 10^{-5}$ to obtain effective feature weights $W_p^{\mathrm{att}}$. Moreover, the channel-token selection calculates the global weights $C_i$ through a global average pooling (GAP) layer and a sigmoid function, followed by masking out channels with weights below $\tau$ to derive channel-wise weights $W_c^{\mathrm{att}}$. The process can be expressed as

$$\begin{cases} P_{\mathrm{ij}} = \mathrm{Softmax}\left(\mathrm{DWConv}\left(F'_L\right)\right) \\ W_p^{\mathrm{att}} = \begin{cases} 0, & \text{if } P_{\mathrm{ij}} < \tau \\ P_{\mathrm{ij}}, & \text{otherwise} \end{cases} \end{cases} \tag{9}$$

$$\begin{cases} C_i = \mathrm{Softmax}\left(\mathrm{GAP}\left(F'_G\right)\right) \\ W_c^{\mathrm{att}} = \begin{cases} 0, & \text{if } C_i < \tau \\ C_i, & \text{otherwise} \end{cases} \end{cases} \tag{10}$$

where $i$ denotes the $i$th channel, and $j$ denotes the $j$th pixel in the $i$th channel. Subsequently, $W_p^{\mathrm{att}}$ and $W_c^{\mathrm{att}}$ are cross-multiplied with $F'_G$ and $F'_L$, respectively, and a learnable parameter $\gamma \in [0, 1]$ is then randomly initialized and optimized during network training to adaptively combine the interaction-enhanced features. These operations are illustrated in Fig. 5, calculated as

$$\hat{F}_1 = \gamma \cdot \left(F'_L \otimes W_p^{\mathrm{att}} \otimes W_c^{\mathrm{att}}\right) + (1-\gamma) \cdot \left(F'_G \otimes W_c^{\mathrm{att}} \otimes W_p^{\mathrm{att}}\right) \tag{11}$$

where $F_1$ is then added to the input $F_0$, and the summed features are passed through an LN layer and a feed-forward neural network (FFN) to obtain the output. This output is subsequently processed by a $3 \times 3$ convolution and undergoes a skip connection with $F_0$ to obtain the first deep feature $F_1$

$$F_1 = \mathrm{Conv}_{3\times3}\left(\mathrm{FFN}\left(\mathrm{LN}\left(F_0 + F_1\right)\right)\right) + F_0. \tag{12}$$

### C. Degradation Estimation

*1) Basic Assumption for Degradation Simulation:* Considering the relationship between spatial and spectral dimensions, we assume that the observed $Y$ can be spectrally degraded from $\hat{X}$ and $Z$ is spatially degraded from $\hat{X}$. The simulation process can be expressed as

$$Y = f_{D_{\mathrm{spec}}}\left(\hat{X}\right), Z = f_{D_{\mathrm{spa}}}\left(\hat{X}\right). \tag{13}$$

To ensure the reliability of degradation simulation, we thoroughly explore the potential consistency characteristics within MSI-OLCI data pairs, which serve as high-quality guidance and critical constraints for degradation learning. By applying spatial degradation from $Y$ and spectral degradation from $Z$, $\widetilde{X}_1 \in \mathbb{R}^{h\times w\times c}$ and $\widetilde{X}_2 \in \mathbb{R}^{h\times w\times c}$ can be derived

$$\tilde{X}_1 = f_{D_{\mathrm{spa}}}\left(Y\right), \tilde{X}_2 = f_{D_{\mathrm{spec}}}\left(Z\right). \tag{14}$$

By combining (13) and (14), we can infer that tilde$X_1$ and $\tilde{X}_2$ should be approximately equal. Based on the above assumption, we define the degradation learning loss term $L_{\mathrm{Deg}}$ as follows:

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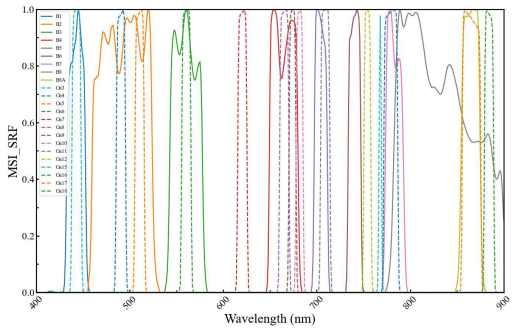



Fig. 6. Comparison of SRF between MSI and OLCI data. Solid and dashed lines represent the SRF of MSI and OLCI data, respectively.

$$L_{\text{Deg}} = \left\| \tilde{X}_1 - \tilde{X}_2 \right\|_1 = \left\| f_{D_{\text{spa}}}(Y) - f_{D_{\text{spec}}}(Z) \right\|_1 \tag{15}$$

where $\|\cdot\|_1$ is a loss function based on the $\ell_1$ norm, which describes the discrepancy between two simulated variables. Once $L_{\text{Deg}}$ optimization is completed, the pretrained spatial degradation function $f_{D_{\text{spa}}}(\cdot)$ is fixed and applied to FATSSR's outputs, ensuring alignment with OLCI data through imposed constraints. Meanwhile, $f_{D_{\text{spec}}}(\cdot)$ spectrally degrades FATSSR's outputs to ensure consistency with MSI data. The details of $f_{D_{\text{spa}}}(\cdot)$ and $f_{D_{\text{spec}}}(\cdot)$ will be detailed in Sections III-C2 and III-C3, respectively.

*2) Pretrained Spatial Degradation Learning:* For spectral reconstruction, existing approaches rely on predefined linear models or simple learnable networks for obtaining spatial and spectral degradation parameters [36], [37]. However, the 15-fold difference in spatial resolution between MSI and OLCI means that simple linear models or average pooling for spatial degradation inherently lose high-frequency details, leading to excessively smoothed outputs.

To address this, we employ the pretrained module to learn the spatial mapping between coarse- and fine-resolution space. For the input $Y$, the upper branch simulates a blurred image by the average pooling layer with an $r \times r$ kernel, and $r$ is the scale factor between spatial grids of $Y$ and $Z$. The lower branch first employs zero-mean normalization (ZmNorm) [38] to eliminate average features while preserving high-frequency components. Subsequently, it enhances these features through DWConv and PReLU activation to maintain textural and edge information. Finally, the two pathways are integrated through element-wise addition followed by a $3 \times 3$ convolution, enhancing the model's ability to handle complex spatial degradation patterns. The process is expressed as follows:

$$\begin{aligned}\tilde{X}_1 = \text{Conv}_{3\times 3}\big(&\text{AvgPool}_{r\times r}(Y) + \text{PReLU} \\ &(\text{DWConv}(\text{ZmNorm}(Y)))\big).\end{aligned} \tag{16}$$

*3) Linear Spectral Degradation Model:* Given the demonstrated high radiometric fidelity and strong interband correlations in MSI and OLCI observations (Fig. 6), we formulate the spectral degradation as a linear weighted model based on their spectral response functions (SRFs) [39], [40]. Specifically, the SRF values of MSI data are used to perform weighted aggregation of the narrow bands that fall within the wide band's wavelength range, thereby achieving spectral degradation from narrow-band to wideband data. The simulation formula is as follows:

$$\tilde{X}_{2,i} = \frac{\sum_{n\in[1,N_i]} Z_n \times \text{SRF}_{\text{MSI}}(\lambda_n)}{\sum_{n\in[1,N_i]} \text{SRF}_{\text{MSI}}(\lambda_n)} \tag{17}$$

where $\tilde{X}_{2,i}$ denotes the $i$th wideband reflectance value after spectral degradation. For the $i$th band of the MSI sensor, there could be more than one corresponding OLCI band. Accordingly, $Z_n$ represents the band reflectance of the $n$th band among the $N_i$ OLCI bands falling within the spectral range of the $i$th MSI band. Moreover, $\lambda_n$ is the central wavelength of the $n$th OLCI band, which is used to determine the spectral response value of MSI's SRF value at $\lambda_n$.

### D. Physics-Informed SA Loss Function

Given that water-leaving $R_{\text{rs}}$ is characterized by weak signals and rich spectral features related to absorbing and scattering effects of water components, we design a SA loss function that incorporates constraints on both IOPs and AOPs of water to achieve precise spectral reconstruction for $R_{\text{rs}}$ over inland waters. IOPs characterize the fundamental light-water interaction physics that are invariant to illumination conditions, whereas AOPs describe the measurable radiative signals that depend on both water constituents and ambient light fields. The synergistic integration of IOPs and AOPs preserves the intrinsic bio-optical characteristics while simultaneously accounting for their actual spectral properties, providing physical constraints for improved accuracy in spectral reconstruction.

*1) IOPs-Physics Driven Loss:* IOPs include parameters such as the total absorption coefficient a, total backscattering coefficient $b_b$, and particulate backscattering coefficient $b_{\text{bp}}$. Among these parameters, $b_b$ demonstrates a direct correlation with satellite-derived backscattering signals and exhibits significantly lower sensitivity to variations in environmental conditions (e.g., meteorological factors and illumination variations) compared to other optical parameters [41]. To leverage these advantages, we incorporate $b_b$ as a physical constraint to facilitate the reconstruction of optical features with improved spectral fidelity. Notably, $b_b$ exhibits superior consistency across wavebands compared to $R_{\text{rs}}$ to characterize the complex optical properties across diverse water types, significantly enhancing the robustness of the model training process. The loss function is formulated as follows:

$$L_{b_b} = \left\| b_b\left(\widehat{R_{\text{rs}}}^{f_{D_{\text{spec}}}(\hat{X})}\right) - b_b\left(\widehat{R_{\text{rs}}}^{Y}\right) \right\|_1 \tag{18}$$

where $\widehat{R_{\text{rs}}}^{Y}$ represents the observed $R_{\text{rs}}$ of MSI data, and $f_{D_{\text{spec}}}(\cdot)$ represents the spectral degradation operation applied to the $\hat{X}$. Specifically, the QAA-750E algorithm is utilized to transform the water $R_{\text{rs}}$ retrieved from the image into $b_b$ for inland water bodies [42]. The $b_b$ calculation formula is as follows:

$$\begin{cases} r_{\text{rs}}(\lambda) & = \frac{R_{\text{rs}}(\lambda)}{0.52+1.7\ R_{\text{rs}}(\lambda)} \\ b_b(\lambda) & = b_{\text{bp}}(\lambda_0)\left(\frac{\lambda_0}{\lambda}\right)^{\left(3.99-3.59\exp\left[-0.9\frac{r_{\text{rs}}(443)}{r_{\text{rs}}(560)}\right]\right)} \\ & \quad + b_{\text{bw}}(\lambda) \end{cases} \tag{19}$$

where $\lambda$ represents the wavelength of $R_{\text{rs}}$ and $\lambda_0 = 740$ nm, and $b_{\text{bw}}$ represents the backscattering coefficient at $\lambda$ for pure

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water. The $b_{\mathrm{bw}}$ can be accessed via the NASA Ocean Color repository: https://www.earthdata.nasa.gov/data

*2) AOPs-Shaped Driven Loss:* AOPs characterize light fields affected by both water constituents and external factors, which generally include $R_{\mathrm{rs}}$, diffuse attenuation coefficient $K_d$, and normalized water-leaving radiance $L_n$. To ensure spectral fidelity, we employ the spectral angle mapper (SAM) loss for $R_{\mathrm{rs}}$, which measures the angular similarity between spectral vectors. This guarantees that reconstructed spectra are well consistent with real observations. The formula is as follows:

$$L_{\mathrm{SAM}} = \alpha \sum_{i=1}^{\mathrm{WH}} \cos^{-1}\left(\widehat{R_{\mathrm{rs}}}^{Z} \cdot \widehat{R_{\mathrm{rs}}}^{f_{D_{\mathrm{spa}}}(\hat{X})} \Big/ \sqrt{\widehat{R_{\mathrm{rs}}}^{Z^{\mathrm{T}}} \cdot \widehat{R_{\mathrm{rs}}}^{Z}} \sqrt{\widehat{R_{\mathrm{rs}}}^{f_{D_{\mathrm{spa}}}(\hat{X})^{\mathrm{T}}} \cdot \widehat{R_{\mathrm{rs}}}^{f_{D_{\mathrm{spa}}}(\hat{X})}}\right) \tag{20}$$

where $\widehat{R_{\mathrm{rs}}}^{Z}$ represents the observed $R_{\mathrm{rs}}$ of OLCI data, and $f_{D_{\mathrm{spa}}}(\cdot)$ represents the spatial degradation operation applied to the $\hat{X}$.

*3) Spatial–Spectral Consistency Loss:* In addition to the strong constraint of water body characteristics, we directly incorporate spectral consistency constraints on the degraded images. Specifically, the widely used $\ell_1$ norm is employed to jointly constrain both the spectral degradation loss and spatial degradation loss

$$L_{\mathrm{SSC}} = \|f_{D_{\mathrm{spa}}}(\hat{X}) - Z\|_1 + \|f_{D_{\mathrm{spec}}}(\hat{X}) - Y\|_1. \tag{21}$$

Finally, the SA loss function can be expressed as follows:

$$L_{\mathrm{SA}} = L_{\mathrm{SSC}} + \lambda L_{b_b} + \mu L_{\mathrm{SAM}} \tag{22}$$

where $\lambda$ and $\mu$ serve as balancing parameters, which are empirically set to 0.65 and 0.001, respectively, in this study. Furthermore, we conducted experimentation on parameter settings for different water types. The suggested hyperparameters for turbid and clean water types are given in the public code package at https://github.com/AQUA-RS-CUG/W-SpectralFormer

## IV. Experiments and Results

### A. Experimental Setup

*1) Performance Evaluation:* Six deep learning-based methods are selected as competitors, including HSCNN+ [17], AWAN [22], SSDCN [43], MST++ [23], MFormer [44], and FSDFF [45]. Among the comparison methods, HSCNN+, AWAN, and MST++ achieved first place in the New Trends in Image Restoration and Enhancement (NTIRE) spectral reconstruction challenges in 2018, 2020, and 2022, respectively. For a fair comparison, we adjusted the training strategy for the supervised methods, including HSCNN+, MST++, and FSDFF, and adopted the weakly supervised two-stage training strategy used in this study. In the training process, the OLCI data are used as the spectral reference, and spatial degradation learning and spectral reconstruction are performed successively. Contrarily, AWAN adopts a weakly supervised training strategy that jointly learns spatial degradation parameters and performs spectral reconstruction, while SSDCN and MFormer are fully unsupervised SSR methods. Due to the lack of ideal reference data with 20-m spatial resolution and narrow spectral bands, we evaluate the reconstruction results from three perspectives.

- Case 1 *Validation of Reconstruction Results Using Real OLCI Data:* All SSR reconstruction results were spatially downsampled to 300 m to match OLCI data resolution, and the spectral quality was assessed by comparing reconstructed spectra against actual OLCI reflectance.
- Case 2 *Validation of Reconstructed Results Using In Situ $R_{\mathrm{rs}}$:* For the reconstruction results of all methods, spatially matched $R_{\mathrm{rs}}$ values were extracted and compared with in situ $R_{\mathrm{rs}}$ values for accuracy assessment.
- Case 3 *Evaluation of the Derived Chla Concentration Using In Situ Measurements:* A random forest (RF) inversion model was trained to obtain the Chla concentration from the reconstructed reflectance data, and the derived Chla values were compared with the in situ Chla values for accuracy evaluation.

The commonly used metrics were used for statistical evaluation of spectral quality, including root-mean-square error (RMSE), structural similarity (SSIM), universal image quality index (UIQI), and SAM. Moreover, $R^2$, RMSE, mean relative error (MRE), and Bias are used to present the accuracy of Chla inversion results. The formulae for these metrics are as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{23}$$

$$\mathrm{SSIM} = \frac{2\mu_{\hat{y}}\mu_y + c_1}{\mu_{\hat{y}}^2 + \mu_y^2 + c_1} \cdot \frac{2\sigma_{\hat{y}y} + c_2}{\sigma_{\hat{y}}^2 + \sigma_y^2 + c_2} \tag{24}$$

$$\mathrm{UIQI} = \frac{\sigma_{\hat{y}y}}{\sigma_{\hat{y}}\sigma_y} \cdot \frac{2\mu_{\hat{y}}\mu_y}{\mu_y^2 + \mu_{\hat{y}}^2} \cdot \frac{2\sigma_{\hat{y}}\sigma_y}{\sigma_y^2 + \sigma_{\hat{y}}^2} \tag{25}$$

$$\mathrm{SAM} = \frac{1}{N}\sum_{i=1}^{N}\cos^{-1}\left(\frac{\sum_{j=1}^{B} y_i^j \hat{y}_i^j}{\sqrt{\sum_{j=1}^{B}\left(y_i^j\right)^2 \sum_{j=1}^{B}\left(\hat{y}_i^j\right)^2}}\right) \tag{26}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{27}$$

$$\mathrm{MRE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i - \hat{y}_i|}{y_i} \tag{28}$$

$$\mathrm{Bias} = 100 \times \mathrm{sign}(Z) \times \left(e^{|Z|} - 1\right) \tag{29}$$

$$Z = \mathrm{median}\left(\log_{10}(\hat{y}_i) - \log_{10}(y_i)\right) \tag{30}$$

where $N$ denotes the total number of pixels in the image. $y_i$ and $\hat{y}_i$ represent the $i$th pixel in the real and predicted image, respectively. $\mu_y$ and $\mu_{\hat{y}}$ denote the mean values of the real and predicted image, respectively. $\bar{y}$ represents the average value of the in-situ samples for accuracy evaluation in Chla inversion tests. $c_1$ and $c_2$ are constants used to maintain stability, and $B$ denotes the number of spectral bands. Moreover, $\sigma_{\hat{y}y}$ denotes the covariance between the predicted and real image, and $\sigma_y$ and $\sigma_{\hat{y}}$ denote the standard deviations of the real and predicted

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TABLE IV

AVERAGE QUANTITATIVE EVALUATION OF SPECTRAL CONSISTENCY WITH SPECTRALLY RECONSTRUCTED RESULTS OBTAINED FROM DIFFERENT METHODS AND OLCI DATA ON LAKES DONGHU, LIANGZIHU, AND CHAOHU. THE CHAMPION AND SECOND PLACE METHODS FOR EACH METRIC ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

| Dataset | Method | | RMSE | SSIM | UIQI | SAM |
|---|---|---|---|---|---|---|
| Lake Donghu | w/o OLCI | SSDCN | 0.0118 | 0.9404 | 0.6406 | 16.163 |
| | | Mformer | 0.0074 | 0.9615 | 0.7864 | 8.5750 |
| | with OLCI | AWAN | 0.0055 | 0.9785 | 0.8651 | 8.7478 |
| | | HSCNN+ | 0.0046 | 0.9829 | 0.8935 | 8.5867 |
| | | MST++ | 0.0047 | 0.9826 | 0.8906 | 8.2661 |
| | | FSDFF | 0.0046 | 0.9832 | 0.8937 | 8.0653 |
| | | Ours | 0.0040 | 0.9896 | 0.9338 | 7.7865 |
| Lake Liangzihu | w/o OLCI | SSDCN | 0.0139 | 0.9564 | 0.7711 | 9.8445 |
| | | Mformer | 0.0082 | 0.9684 | 0.8400 | 8.1180 |
| | with OLCI | AWAN | 0.0040 | 0.9872 | 0.9275 | 7.2916 |
| | | HSCNN+ | 0.0034 | 0.9893 | 0.9366 | 7.2134 |
| | | MST++ | 0.0036 | 0.9882 | 0.9320 | 7.2875 |
| | | FSDFF | 0.0031 | 0.9906 | 0.9419 | 7.3233 |
| | | Ours | 0.0026 | 0.9943 | 0.9636 | 6.9009 |
| Lake Chaohu | w/o OLCI | SSDCN | 0.0205 | 0.9239 | 0.7286 | 9.1522 |
| | | Mformer | 0.0134 | 0.9453 | 0.8258 | 6.7225 |
| | with OLCI | AWAN | 0.0069 | 0.9750 | 0.9087 | 6.1545 |
| | | HSCNN+ | 0.0067 | 0.9791 | 0.9221 | 5.9046 |
| | | MST++ | 0.0069 | 0.9798 | 0.9236 | 6.0209 |
| | | FSDFF | 0.0061 | 0.9833 | 0.9341 | 6.0999 |
| | | Ours | 0.0048 | 0.9897 | 0.9555 | 5.3896 |

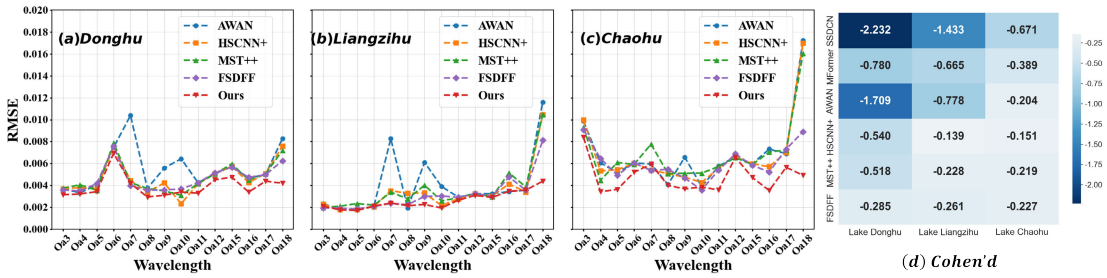



Fig. 7. RMSE values for each band of the spectrally reconstructed results obtained from different methods across three lakes, where OLCI spectral bands are used as the reference. (a)–(c) Results for Lake Donghu, Lake Liangzihu, and Lake Chaohu, respectively. (d) Cohen's d values of the SAM metric differences for our method compared to others across the three lakes.

image, respectively. Larger values of SSIM, UIQI, $R^2$, and smaller values of RMSE, SAM, MRE, and Bias indicate better reconstruction quality.

*2) Implementation Details:* The pretrained spatial degradation learning (P-SDL) module was trained for 60 epochs using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$) with an initial learning rate of $5 \times 10^{-4}$, halved after five stagnant epochs, and gradient clipping with a threshold of 0.1 was used for stability. The FATSSR module was trained for 100 epochs using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$) with an initial learning rate of $2 \times 10^{-4}$, halved after five stagnant epochs. The patch sizes for MSI and OLCI inputs were set to $128 \times 128$ and $9 \times 9$, respectively, and the number of FATM blocks was $N = 2$. For the comparison methods, the parameters were set to their default values and retrained with the same datasets used in this study (see Table III). In addition, a dataset of 403 in situ hyperspectral $R_{rs}$ measurements with synchronized Chla concentrations, collected across the middle and lower reaches of the Yangtze River, was used to train the RF model for Chla inversion. In this dataset, 300 samples were allocated for model training, while 103 samples were used for accuracy testing [46]. Specificall, the input feature set of the RF model includes all reconstructed spectral bands, along with the Normalized Fluorescence Line Height index (FLH, $R_{rs}(681)/R_{rs}(665)$) and the band ratio ($R_{rs}(704)/R_{rs}(674)$) [47]. For Chla estimation, the algal bloom area was identified using both the normalized difference vegetation index (NDVI) with a threshold >0 and the floating algae index (FAI) with a threshold > −0.004 [48], [49], and two pixels adjacent to the shorelines at 20-m resolution were removed from the Chla map.

## B. Experimental Results

*1) Result Validation Using Real OLCI Data:* First, we compare the reconstruction results with the real OLCI data to evaluate the spectral quality, as illustrated in Section IV-A1. Table IV shows the average spectral differences between the spatially downsampled reconstruction results and OLCI data across the 14 bands for each of the three study sites. The results show that the two unsupervised methods (i.e., SSDCN and MFormer) without using OLCI spectral information achieve the poorest performance across various lakes, especially in terms of RMSE and SAM metrics. This indicates that unsupervised SSR methods can lead to significant spectral

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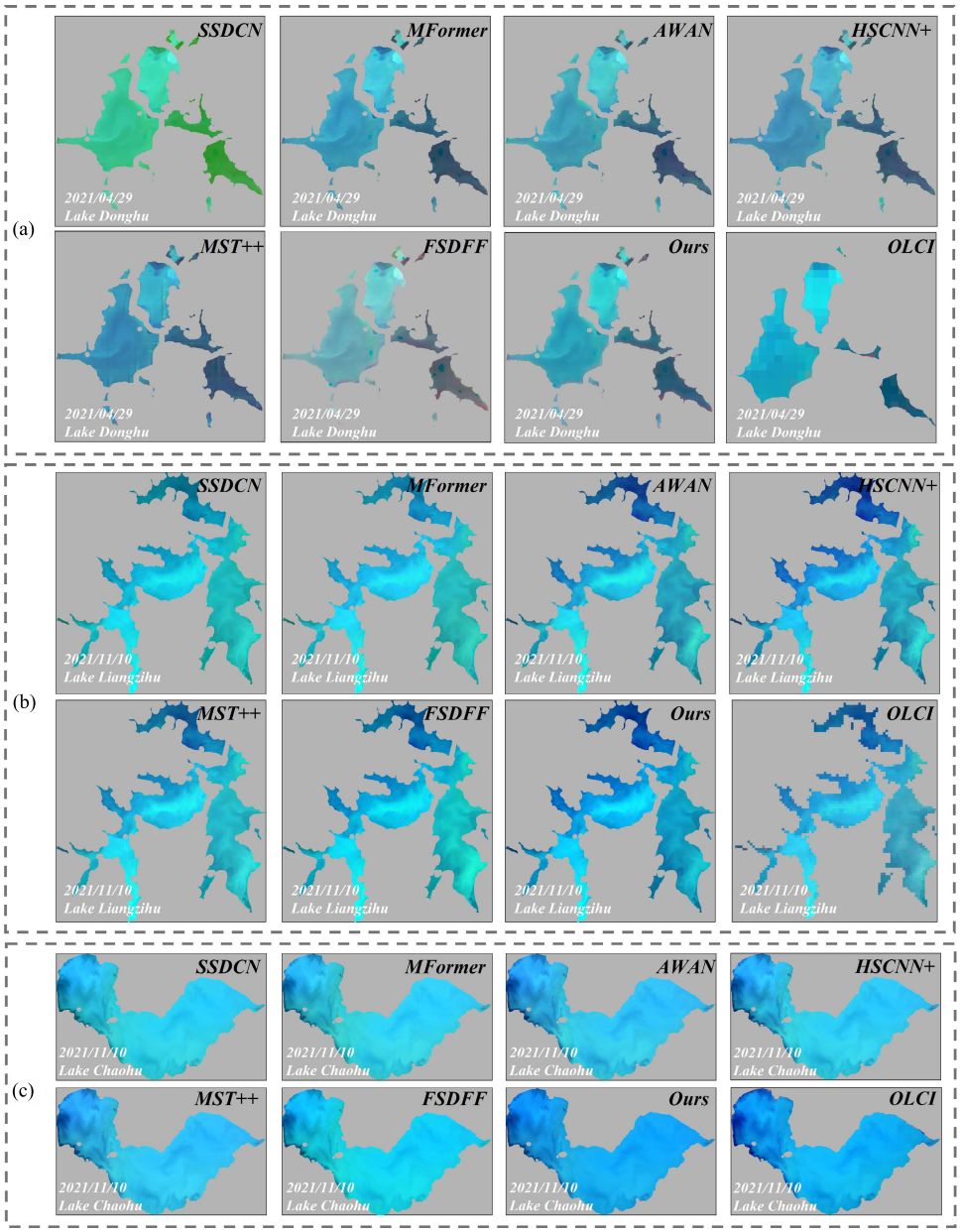



Fig. 8. Visual results obtained by different SSR methods for (a) Lake Donghu, (b) Lake Liangzihu, and (c) Lake Chaohu, respectively. All images are displayed in false-color (Oa18–Oa9–Oa5) representation.

distortion, which is unsuitable for the monitoring of water environments that require high spectral quality. Comparatively, the other methods in the manner of weakly supervised learning achieve more reliable results in the tests. In the three study sites, the proposed W-SpectralFormer method achieves the best accuracy in terms of all the evaluation metrics.

For the comparison methods, FSDFF achieves the suboptimal performance by effectively integrating the global information and local structural features. Both MST++ (employing a deep recursive U-Net framework) and HSCNN+ (utilizing a lightweight CNN architecture) achieve comparable reconstruction accuracy, which suggests that excessive layer stacking in neural networks may not be a good strategy for model optimization for aquatic applications. Although AWAN incorporates a CAM into its CNN-based architecture to adaptively adjust channel-wise feature weights, it fails to deliver competitive performance in the tests. This shows that in weakly supervised SSR, both degradation parameter estimation and network architecture significantly influence the fidelity of reconstructed spectra.

Compared with these baseline models, W-SpectralFormer demonstrates superior performance across all three lakes due to its ability to adaptively exploit inherent spectral characteristics in different lakes. Based on the results in Table IV, W-SpectralFormer achieves an average SAM of 6.6923 across all lakes, representing a 6.57% reduction in spectral distortion compared to FSDFF (i.e., 7.1628 with the second-best performance). Specifically, W-SpectralFormer achieves a 13.04% lower RMSE and a 3.46% smaller SAM compared to FSDFF in Lake Donghu. The improvements are even more pronounced in Lake Liangzihu, where W-SpectralFormer reduces RMSE by 16.13% relative to FSDFF and by 23.53% compared to

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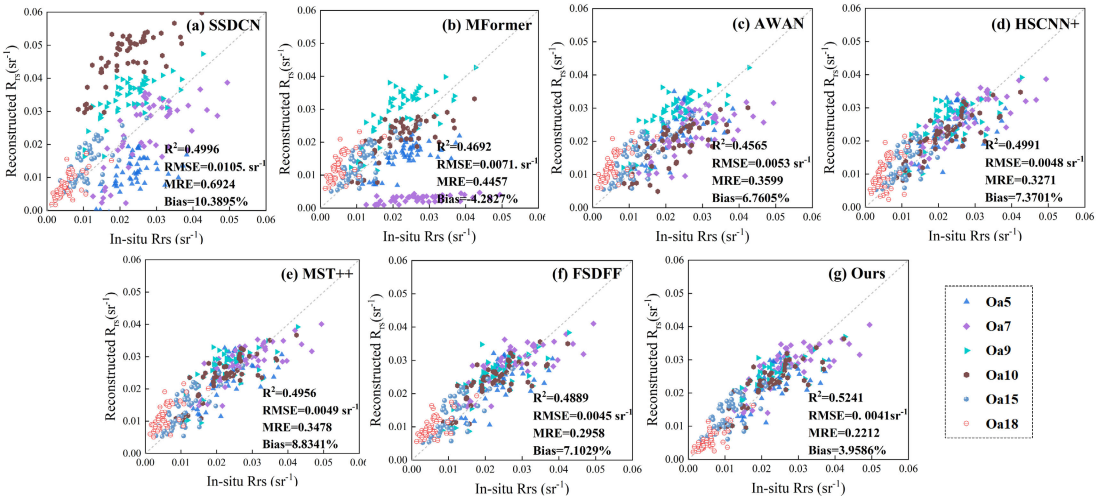



Fig. 9. Scatter plots of reconstructed $R_{rs}$ and 54 in situ matchups across three lakes. (a)–(g) Performance of different SSR methods for six reconstructed bands (i.e., Oa5, Oa7, Oa9, Oa10, Oa15, and Oa18), respectively.

HSCNN+, respectively. Furthermore, we evaluate the reconstruction accuracy across all 14 spectral bands by comparing RMSE values among different methods. As illustrated in Fig. 7, W-SpectralFormer achieves the lowest RMSE values in the majority of spectral bands, demonstrating its superior capability for high-fidelity spectral image reconstruction. Moreover, Cohen's d indicates that our method, compared to others, exhibits a meaningful and practical difference in the SAM metric across the three lakes.

To better illustrate the spatial and spectral reconstruction performance of different methods, we visually compare the reconstructed images with the synchronized OLCI images as the spectral reference. In Fig. 8, we employ three reconstructed bands (i.e., Oa5, Oa9, and Oa18) for false-color representation. Overall, SSDCN and MFormer exhibit significant deviations in water colors across all three lakes. HSCNN+ produces reconstructed images with noticeable color discontinuities near the shoreline in Lake Donghu, suggesting inaccurate spectral feature modeling. Meanwhile, MST++ consistently generates artifacts across all test lakes, revealing limitations in its generalization ability when applied to aquatic environments despite its complex architecture. While AWAN and FSDFF demonstrate enhanced spatial detail reconstruction, their results suffer from observable spectral distortions in Lakes Donghu and Chaohu, as well as anomalies in brighter regions of Lake Liangzihu. In contrast, the proposed W-SpectralFormer exhibits minimal spectral deviation from OLCI reference data in the three lakes. Both quantitative and qualitative analyses confirm that W-SpectralFormer outperforms alternative methods across all three test lakes, demonstrating robust spectral reconstruction capabilities.

*2) Result Validation Using In Situ $R_{rs}$:* Table V summarizes the average spectral reconstruction accuracy by comparing the reconstructed reflectance and the matched in situ spectra across all bands for each lake. Consistent with the analysis results in Section IV-B1, the unsupervised methods SSDCN and MFormer still achieve the highest RMSE values, indicating their poorest performance. The proposed W-SpectralFormer achieves the lowest average RMSE of 0.0041 $sr^{-1}$ across all three lakes, representing an 8.89% accuracy improvement over 0.0045 $sr^{-1}$ of FSDFF. Fig. 9, visually compares the performance of different methods across the six reconstructed bands (Oa5, Oa7, Oa9, Oa10, Oa15, and Oa18) at all matchups. Overall, W-SpectralFormer achieves optimal spectral fidelity between reconstructed and in situ spectra (Fig. 9(g)), demonstrating superior quantitative performance ($R^2$ = 0.5241 and RMSE = 0.0041 $sr^{-1}$). By contrast, SSDCN and MFormer show substantial deviations between reconstructed and in situ spectra (Fig. 9(a) and (b)). Among the other comparison methods, HSCNN+, MST++, and FSDFF demonstrate superior

TABLE V

MEAN QUANTITATIVE EVALUATION OF RECONSTRUCTED AND IN SITU SPECTRA ACROSS 14 BANDS [LAKE DONGHU ($N$ = 16), LAKE LIANGZIHU ($N$ = 17), AND LAKE CHAOHU ($N$ = 21)]. BEST IS IN RED AND SECOND BEST IN BLUE

| Dataset | Method | | $R^2$ | RMSE | MRE | Bias (%) |
|---|---|---|---|---|---|---|
| Lake Donghu | w/o OLCI | SSDCN | 0.2768 | 0.0098 | 0.7322 | 3.8218 |
| | | Mformer | 0.2762 | 0.0060 | 0.4041 | -12.8521 |
| | with OLCI | AWAN | 0.2707 | 0.0047 | 0.3292 | 0.1898 |
| | | HSCNN+ | 0.2834 | 0.0042 | 0.2633 | -0.1088 |
| | | MST++ | 0.2738 | 0.0043 | 0.3111 | 3.3779 |
| | | FSDFF | 0.2786 | 0.0042 | 0.2987 | 3.3992 |
| | | Ours | 0.3331 | 0.0038 | 0.2089 | -0.2677 |
| Lake Liangzihu | w/o OLCI | SSDCN | 0.5824 | 0.0105 | 0.5917 | 8.1498 |
| | | Mformer | 0.5355 | 0.0076 | 0.3838 | -4.9522 |
| | with OLCI | AWAN | 0.5700 | 0.0053 | 0.2950 | 5.3228 |
| | | HSCNN+ | 0.5707 | 0.0047 | 0.2706 | 6.1167 |
| | | MST++ | 0.5826 | 0.0047 | 0.2769 | 6.9839 |
| | | FSDFF | 0.5651 | 0.0045 | 0.2555 | 5.6935 |
| | | Ours | 0.5888 | 0.0042 | 0.1836 | 2.2696 |
| Lake Chaohu | w/o OLCI | SSDCN | 0.6397 | 0.0112 | 0.7533 | 19.1968 |
| | | Mformer | 0.5959 | 0.0077 | 0.5493 | 4.9563 |
| | with OLCI | AWAN | 0.5289 | 0.0060 | 0.4554 | 14.7688 |
| | | HSCNN+ | 0.6433 | 0.0056 | 0.4475 | 16.1024 |
| | | MST++ | 0.6304 | 0.0056 | 0.4553 | 16.1406 |
| | | FSDFF | 0.6229 | 0.0047 | 0.3333 | 12.2161 |
| | | Ours | 0.6504 | 0.0042 | 0.2711 | 9.8740 |

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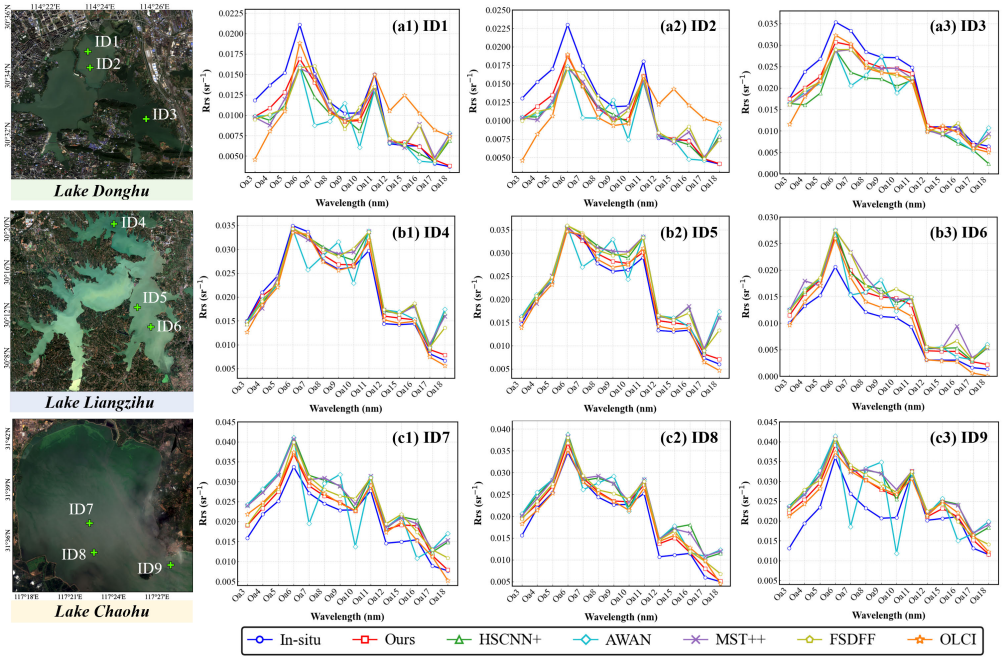



Fig. 10. Spectral curves of different SSR methods and in situ matchups at selected sampling points. (a1)–(a3) Comparative results for Lake Donghu, while (b1)–(b3) and (c1)–(c3) comparison for Lake Liangzihu and Lake Chaohu, respectively.

reconstruction accuracy over AWAN, which implies the critical importance of training strategy in SSR tasks.

In low-reflectance bands (Oa15 and Oa18) shown in Fig. 9, conventional SSR methods struggle to preserve lake spectral characteristics accurately. In those cases, W-SpectralFormer still reconstructs faithful spectral signatures that closely match the true water spectra. This superior performance can be attributed to the incorporation of constraints to IOPs, which converts low-magnitude $R_{rs}$ into $b_b$, thereby normalizing the dynamic range across spectral bands.

In Fig. 10, the reconstructed spectra of different methods at each site are compared with the in situ spectra. To better demonstrate the detailed differences in spectral reconstruction of other methods, we exclude the SSDCN and MFormer methods that have poor quantitative accuracies, as shown in Table V and Fig. 9. For all the sample points, W-SpectralFormer shows the optimal consistency with the in situ $R_{rs}$ and OLCI data. In addition, W-SpectralFormer exhibits superior accuracy in the reconstruction bands of Oa9–Oa12 for ID5-6 in Lake Liangzihu, and ID7-8 in Lake Chaohu. These bands within 670–710-nm record characteristic Chla fluorescence features that are essential for robust Chla concentration retrieval in aquatic environments. Through explicit constraint of $b_b$, W-SpectralFormer effectively prioritizes chlorophyll-sensitive spectral bands during reconstruction by adaptively adjusting their gradient weights. This enables robust preservation of lake-specific optical properties across diverse aquatic environments and accurate retrieval of Chla concentration.

*3) Evaluation of the Derived Chla Concentration:* In this section, we evaluate the performance of Chla obtained from reconstructed reflectance data of different SSR methods compared to in situ Chla measurements. As introduced in Section IV-A2, a total of 103 test samples distributed over the lakes in the middle and lower reaches of the Yangtze River basin (Fig. 1(d)) are selected for validating the accuracy of the Chla maps derived from the synchronic MSI images. Table VI shows the average quantitative scores of Chla derived from different reconstructed $R_{rs}$ with 20-m spatial resolution and 14 spectral bands. It can be observed that SSDCN and MFormer exhibit poor performance with significantly distorted spectra for quantitative applications, which is consistent with the results in Sections IV-B1 and IV-B2. Among the five weakly supervised learning methods, the proposed W-SpectralFormer achieves the best results ($R^2$ = 0.8240 and RMSE = 13.4660 $\mu$g/L), while HSCNN+ achieves the second-best performance ($R^2$ = 0.8028 and RMSE = 14.4995 $\mu$g/L). The spectral distortion of AWAN in Chla sensitive bands (i.e., Oa7, Oa8, and Oa9) has a certain impact on the inversion accuracy of Chla.

Fig. 11 shows the Chla inversion maps of the study lakes based on different reconstruction results, where the in situ samples are also visualized on the maps. Both SSDCN and MFormer exhibit inaccurate Chla predictions across all lakes, particularly in high-Chla concentration areas (Chla > 70 $\mu$g/L). Compared to other supervised learning methods, W-SpectralFormer achieved more reliable Chla estimates, as

TABLE VI

QUANTITATIVE EVALUATION RESULTS OF THE CHLA DERIVED FROM THE RECONSTRUCTED RESULTS OF DIFFERENT SSR METHODS USING THE IN SITU CHLA ($N$ = 103). BEST IS IN RED AND SECOND BEST IN BLUE

| Method | | $R^2$ | RMSE($sr^{-1}$) | MRE | Bias(%) |
|---|---|---|---|---|---|
| w/o OLCI | SSDCN | 0.1740 | 54.1542 | 2.7329 | 22.7849 |
| | Mformer | 0.7711 | 22.7556 | 0.8225 | 13.8638 |
| with OLCI | AWAN | 0.7954 | 14.9687 | 0.4933 | 1.7931 |
| | HSCNN+ | 0.8028 | 14.4995 | 0.3968 | -3.5057 |
| | MST++ | 0.7834 | 16.2295 | 0.4118 | -6.7555 |
| | FSDFF | 0.7343 | 16.4622 | 0.4252 | -0.3247 |
| | Ours | 0.8240 | 13.4660 | 0.3862 | 2.5434 |

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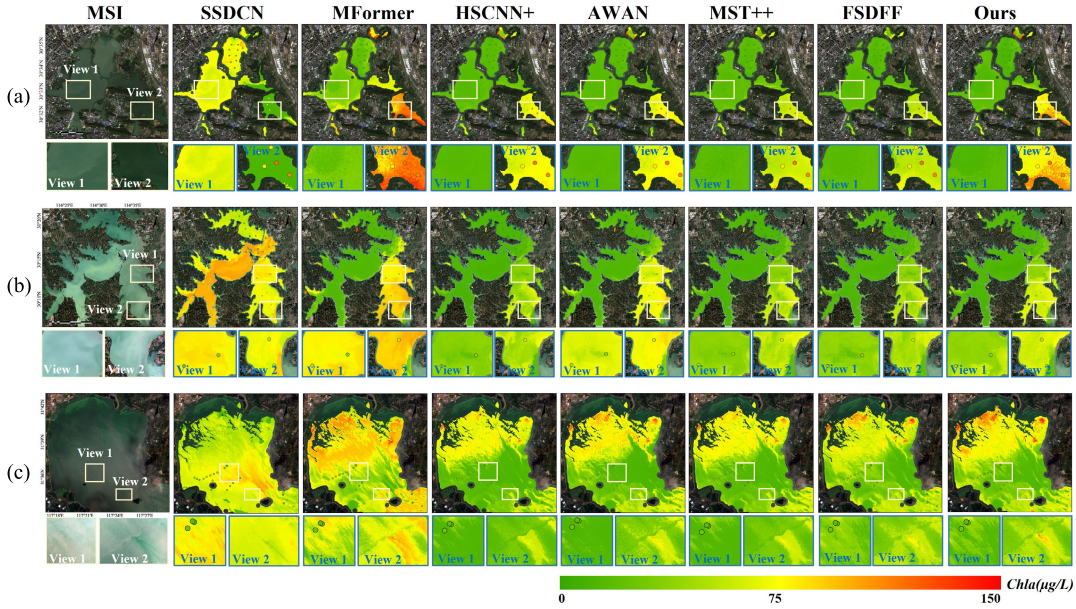



Fig. 11. Chla inversion maps of three lakes based on $R_{rs}$ reconstructed from different SSR methods. (a)–(c) Chla maps for Lakes Donghu, Liangzihu, and Chaohu, respectively. Zoomed-in views of the highlighted regions are provided for clearer visualization. The in situ Chla values are overlaid on the maps using the same color scale as the inversion results.

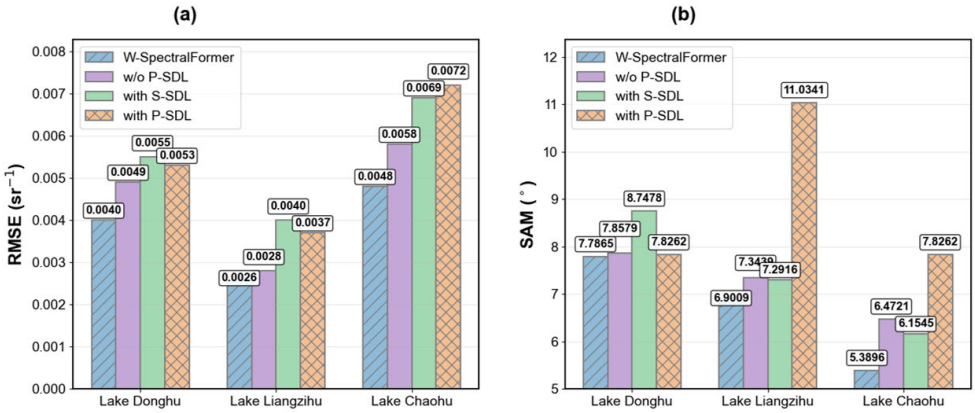



Fig. 12. Impact of P-SDL on W-SpectralFormer performance. (a) RMSE comparison. (b) SAM comparison.

competing approaches often yield underestimated retrievals (as shown in View 2 in Fig. 11(a) and (c)). The zoomed-in-view visualization of three lakes (as shown in View 1 in Fig. 11(b) and (c), and View 2 in Fig. 11(a)) also shows that only W-SpectralFormer-derived Chla concentration maps have with similar textural distribution to real MSI images. The visual analysis demonstrates the model's ability to simultaneously reconstruct high-resolution spectral features while maintaining accurate spatial details, confirming its strong potential for precise water quality monitoring.

## C. Ablation Study

A comprehensive ablation study is conducted to assess the proposed W-SpectralFormer framework, focusing on evaluating the impact of critical network modules and loss functions. The ablation study mainly focuses on quantitatively evaluating the spectral consistency between the reconstructed data and the real OLCI data, as illustrated in Section IV-A1. All reported quantitative metrics represent temporally averaged values across all test dates for each lake.

*1) Effectiveness of P-SDL:* The CNN-based pretrained degradation aims to learn the spatial mapping relationships between MSI-OLCI image pairs. To evaluate its impact on the performance of W-SpectralFormer, this experiment conducts the bicubic downsampling interpolation (labeled as "w/o P-SDL"), simultaneous learning of degradation parameters and SSR (labeled as "with S-SDL"), and the pretrained degradation estimator embedded in our framework (labeled as "with P-SDL") without full convergence to optimal estimation (i.e., epoch = 1 for P-SDL). The experimental results demonstrate that W-SpectralFormer achieves optimal performance across all three study lakes when the full architecture with integrated degradation modeling is employed (Fig. 12). Notably, substituting the proposed P-SDL with either simultaneous learning of degradation parameters and SSR or the P-SDL without full convergence to optimal estimation leads to a significant performance degradation. This empirical comparison validates the

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TABLE VII

ABLATION STUDY ON KEY MODULES IN W-SPECTRALFORMER FRAMEWORK. BEST RESULTS ARE MARKED IN RED, WITH VALUES IN PARENTHESES INDICATING THE PERCENTAGE IMPROVEMENT RELATIVE TO THE BASELINE METRIC IN THE ROW ABOVE

| DConv-FEM | F-token | LG-FAM | Lake Donghu | | Lake Liangzihu | | Lake Chaohu | |
|---|---|---|---|---|---|---|---|---|
| | | | RMSE | SAM | RMSE | SAM | RMSE | SAM |
| × | × | × | 0.0044 | 8.2111 | 0.0044 | 7.4831 | 0.0082 | 6.2743 |
| ✓ | × | × | 0.0041 (-6.82%) | 8.1826 (-0.35%) | 0.0041 (-6.82%) | 7.4166 (-0.89%) | 0.0078 (-4.88%) | 6.1940 (-1.28%) |
| ✓ | ✓ | × | 0.0040 (-2.44%) | 8.0948 (-1.07%) | 0.0035 (-14.63%) | 7.2111 (-2.77%) | 0.0060 (-23.08%) | 5.1793 (-3.04%) |
| ✓ | ✓ | ✓ | 0.0040 (0.00%) | 7.7865 (-3.81%) | 0.0027 (-22.86%) | 6.9009 (-4.30%) | 0.0048 (-20.00%) | 5.3896 (-10.27%) |

TABLE VIII

ABLATION STUDY FOR KEY LOSS FUNCTIONS IN W-SPECTRALFORMER FRAMEWORK. BEST RESULTS ARE MARKED IN RED, WITH VALUES IN PARENTHESES INDICATING THE PERCENTAGE IMPROVEMENT RELATIVE TO THE BASELINE METRIC IN THE ROW ABOVE

| $L_1+L_2$ | $L_{SAM}$ | $L_{b_b}$ | Lake Donghu | | Lake Liangzihu | | Lake Chaohu | |
|---|---|---|---|---|---|---|---|---|
| | | | RMSE | SAM | RMSE | SAM | RMSE | SAM |
| ✓ | × | × | 0.0043 | 8.5777 | 0.0046 | 7.3961 | 0.0082 | 5.9916 |
| ✓ | ✓ | × | 0.0042 (-2.33) | 8.5500 (-0.32%) | 0.0044 (-4.35%) | 7.3380 (-0.79%) | 0.0072 (-12.20%) | 5.7680 (-3.73%) |
| ✓ | ✓ | ✓ | 0.0040 (-4.76%) | 7.7865 (-8.93%) | 0.0027 (-38.64%) | 6.9009 (-5.96%) | 0.0048 (-33.33%) | 5.3896 (-6.56%) |

stability and effectiveness of our proposed spatial degradation estimation strategy.

*2) Effectiveness of Modules Within FATSSR Network:* The original transformer model is used as the baseline, and the key modules (i.e., DConv-FEM, F-token, and LG-FAM) are gradually added into the baseline model to evaluate their effectiveness on the model performance. The accuracy scores for different modules are shown in Table VII. Compared to DConv-FEM, the F-token significantly enhances W-SpectralFormer's performance through adaptive frequency decomposition to better capture fine-grained spectral details. This improvement is particularly pronounced in Lake Chaohu, where frequency-aware processing of F-token demonstrates superior performance in preserving peak and valley spectral features. Meanwhile, the LG-FAM dynamically adjusts global and local feature weights through flexible cross-interactions, also yielding considerable improvement in quantitative accuracies. By integrating these three modules, the proposed method achieves the highest performance scores across all three study lakes.

*3) Effects of Loss Functions:* We further explored the influence of $L_{b_b}$ and $L_{\text{SAM}}$ in the designed SA loss functions, as shown in Table VIII. The results indicate that the model performance improved after introducing these two auxiliary loss terms. Specifically, incorporating the $L_{\text{SAM}}$ loss reduced RMSE by 2.33%, 4.35%, and 12.20% for Lakes Donghu, Liangzihu, and Chaohu, respectively. The $L_{b_b}$ loss achieved greater improvements, with RMSE decreases of 4.76%, 38.64%, and 33.33% for the respective lakes. These results clearly demonstrate the effectiveness of the proposed SA loss function in significantly enhancing model performance.

## V. DISCUSSIONS

### A. Necessity of SSR for Monitoring Inland Waters

In Section IV, we have conducted a comprehensive analysis of the reconstructed results to validate the reliability of the proposed W-SpectralFormer method for spectral reconstruction. However, is it really necessary to conduct SSR for MSI data for monitoring of WQIs? To discuss the advantage of SSR data with high spatial and spectral resolution, we give the analysis for comparing the inversion accuracy of the reconstructed data with real OLCI and MSI data. Fig. 13 shows the quantitative scatter relationships between the in situ Chla and the Chla derived from the MSI, OLCI, and SSR data, respectively. Moreover, Fig. 14 presents a visual comparison of Chla inversion for the three datasets in the study lakes.

The scatter plot of MSI data reveals good performance in low-Chla concentration (i.e., < 50 $\mu$g/L). However, the accuracy degrades for predicting high-Chla values. In Fig. 14(e), the results also show a noticeable overestimation of MSI-derived results in the central part of Lake Liangzihu. This observation suggests that the broad spectral bands of MSI data lack the necessary spectral refinement to effectively capture the sensitive spectral signatures of high-Chla concentrations. In contrast, OLCI data exhibit significant underestimation within the 80–120 $\mu$g/L range, as shown in Fig. 13(a). As these points are mainly distributed in Lake Donghu, characterized by a relatively small water area (Fig. 14(a)), the observed discrepancies primarily result from OLCI's 300-m resolution limitations in resolving Chla heterogeneity and possible land adjacency effects. With 20-m spatial resolution and reconstructed spectral bands, the SSR data can generate a Chla map with good consistency with in situ measurements, and achieved better quantitative accuracies ($R^2$ = 0.8471 and RMSE = 13.4667 $\mu$g/L) than original MSI and OLCI data.

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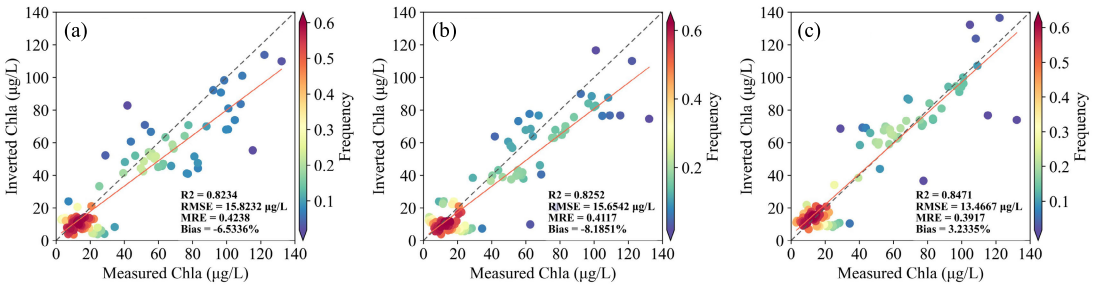



Fig. 13. Scatter plots of satellite-derived Chla maps and 103 in situ matchups. (a)–(c) Scatter plots of inversion results derived from OLCI, MSI, and reconstructed images, respectively. The color gradient of scatter points reflects the agreement between the inverted and in situ values.

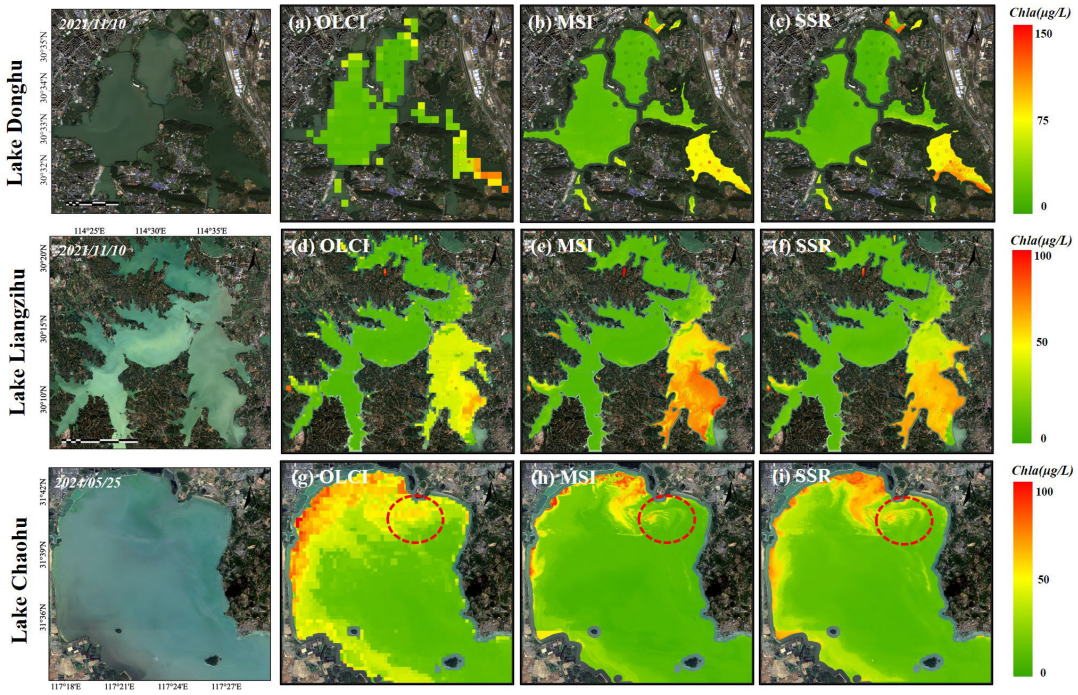



Fig. 14. Chla inversion maps of three lakes based on OLCI, MSI, and SSR images. Specifically, (a)–(c) Chla maps for Lake Donghu, while (d)–(f) and (g)–(i) Chla results for Lake Liangzihu and Lake Chaohu, respectively. The in situ Chla values are overlaid on the maps using the same color scale as the inversion results.

As shown in the highlighted areas in Fig. 14(g)–(i), the SSR-derived Chla map in Chaohu Lake exhibits enhanced spatial coherence with reference water textures. Both quantitative metrics and visual comparisons demonstrate that high spatial resolution and narrowband spectral data are essential for Chla estimation, which further validates the necessity and advantages of the proposed W-SpectralFormer method.

### B. Potential and Limitations

To demonstrate the potential of the proposed W-SpectralFormer method for practical application, we collect the available cloud-free MSI images covering Lakes Donghu, Liangzihu, and Chaohu from September 2023 to September 2024, and use the well-trained W-SpectralFormer model to generate a time series of reconstructed images with a spatial resolution of 20 m and 14 spectral bands. Based on the well-trained RF inversion model, the Chla maps are then obtained for time-series monitoring of the lake environment, as shown in Figs. 15–17. The two urban lakes in Wuhan (Donghu and

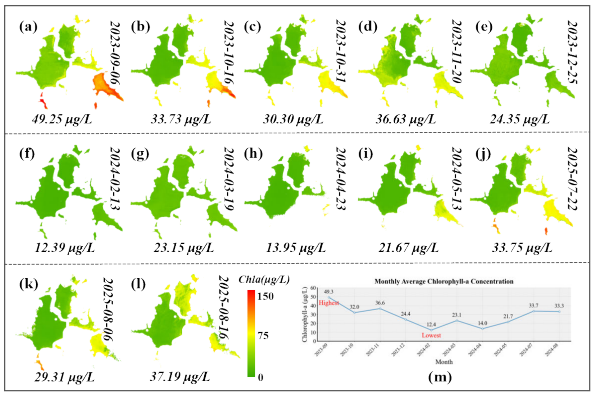



Fig. 15. Chla inversion results of Lake Donghu from September 2023 to September 2024, based on reconstructed MSI images. (a)–(l) correspond to Chla maps derived from each corresponding image and (m) depicts the time series of the mean Chla concentration for the entire lake.

Liangzihu) exhibit similar seasonal Chla dynamics, with minimum concentrations occurring in winter (December–February,

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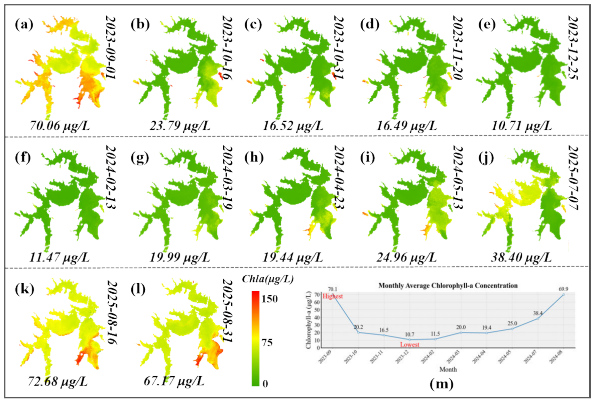



Fig. 16. Chla inversion results of Lake Liangzihu from September 2023 to September 2024, based on reconstructed MSI images. (a)–(l) correspond to Chla maps derived from each corresponding image and (m) depicts the time series of the mean Chla concentration for the entire lake.

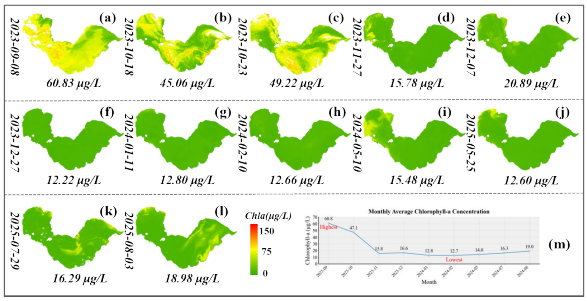



Fig. 17. Chla inversion results of Lake Chaohu from September 2023 to September 2024, based on reconstructed MSI images. (a)–(l) correspond to Chla maps derived from each corresponding image and (m) depicts the time series of the mean Chla concentration for the entire lake.

Figs. 15(m) and 16(m)) and peak levels observed during summer to early autumn (June–September, Fig. 17(m)). In contrast, Lake Chaohu exhibits a peak in Chla concentration during the whole autumn while showing relatively lower levels in other months, aligning with findings from previous studies [50].

With a 20-m spatial resolution, clear spatial details can be observed in the Chla maps. Although two pixels adjacent to the shorelines are excluded in the maps to mitigate the bordering effect of land areas, the results show that high-Chla concentrations tend to occur in the nearshore areas of the lakes. During the warm seasons, the Chla concentrations in the southeastern regions of Lakes Liangzihu and Donghu are higher compared to other areas. In autumn (e.g., September and October), the entire area of Lake Chaohu exhibits a high risk of eutrophication. Overall, the Chla maps derived from reflectance data constructed by W-SpectralFormer can accurately capture the spatial and temporal changes of Chla distribution in inland lakes, which provides strong data support for water quality prevention and early warning.

Although the proposed W-SpectralFormer demonstrates optimal reconstruction performance, its accuracy remains limited in highly eutrophic lakes. In Lake Chaohu, the in situ spectra in Fig. 10(c3) exhibit distinct dual-peak characteristics typical of eutrophic waters. However, W-SpectralFormer overestimates reflectance in visible bands (especially Oa6–Oa9), weakening the dual-peak signature that typically indicates high-Chla concentrations. The spectral distortion could introduce systematic biases in Chla concentration estimates from the RF-based inversion model. Future improvements will investigate integrating optical water type classification coupled with dynamic feature weighting to improve model generalization. Moreover, we propose to use SSR to generate data with high spatial and spectral resolution by leveraging the complementary information from MSI and OLCI data in the spectral and spatial dimensions. However, the exploration of integrating temporal dependencies from multisource data can further promote the imaging density for efficient water environment monitoring. Future work will focus on developing an integrated spatio-spectral–temporal framework to better capture temporal dynamics from OLCI data, enhancing dynamic monitoring capabilities.

TABLE IX
MODEL EFFICIENCY COMPARISON IN TERMS OF FPS AND PARAMETERS

| Method | FPS ↑ | Parameters (MB) ↓ |
|---|---|---|
| SSDCN | 351.52 | 1.58 |
| Mformer | 62.27 | 6.55 |
| AWAN | 148.61 | 15.38 |
| HSCNN+ | 110.69 | 17.66 |
| MST++ | 73.82 | 1.32 |
| FSDFF | 25.67 | 42.35 |
| Ours | 127.37 | 1.85 |

### C. Complexity Analysis

Table IX presents the frames per second (FPS) and parameter size of different deep learning methods on three lakes. All the codes were run on an NVIDIA GeForce RTX 4090 GPU and a 13th Gen Intel Core i7-13700F CPU. The patch size for the computational complexity experiments was set to $128 \times 128$ for a fair comparison of FPS values.

As illustrated in the model efficiency comparison, our method achieves a favorable balance between model size and inference speed. With one of the lowest parameter counts among all compared models, it offers a lightweight architecture ideal for deployment in resource-constrained environments. Meanwhile, it maintains competitive FPS, outperforming several larger models such as HSCNN+ and AWAN. This highlights the efficiency and practicality of our design, which ensures fast inference without sacrificing model compactness.

## VI. CONCLUSION

In this article, we have proposed a physics-informed SSR framework for the SSR of remote sensing reflectance over inland waters. The main contributions of this work include a FATSSR network for high-fidelity spectral reconstruction, a spatial–spectral degradation estimator enabling weakly supervised optimization, and an optical property-driven SA loss for accurate remote sensing reflectance recovery. Experimental validation across three representative inland lakes demonstrates that W-SpectralFormer achieves superior spectral fidelity relative to state-of-the-art SSR methods, as

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evidenced by improved quantitative metrics and visual quality. Comparative analysis against unsupervised approaches confirms that incorporating OLCI data significantly enhances both accuracy and reliability in aquatic SSR applications. Specifically, the FFT-assisted FATSSR network improves reconstruction of fine spectral details while enabling higher resolution Chla texture inversion. Furthermore, the proposed SA loss function facilitates high-fidelity reconstruction of low-reflectance and Chla-sensitive spectral features, which is critical for precise Chla retrieval. This work provides a cost-effective and computationally efficient solution for water reflectance SSR, offering a practical framework for applications in water quality monitoring and aquatic ecosystem studies.

## REFERENCES


[1] A. J. Reid et al., "Emerging threats and persistent conservation challenges for freshwater biodiversity," *Biol. Rev.*, vol. 94, no. 3, pp. 849–873, 2018.

[2] J. Ma, S. Jin, J. Li, Y. He, and W. Shang, "Spatio-temporal variations and driving forces of harmful algal Blooms in chaohu lake: A multi-source remote sensing approach," *Remote Sens.*, vol. 13, no. 3, p. 427, Jan. 2021.

[3] Q. Cao, G. Yu, and Z. Qiao, "Application and recent progress of inland water monitoring using remote sensing techniques," *Environ. Monitor. Assessment*, vol. 195, no. 1, p. 125, Jan. 2023.

[4] D. Kumar, R. Kumar, M. Sharma, A. Awasthi, and M. Kumar, "Global water quality indices: Development, implications, and limitations," *Total Environ. Adv.*, vol. 9, Mar. 2023, Art. no. 200095.

[5] F. Zeng et al., "Monitoring inland water via sentinel satellite constellation: A review and perspective," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 340–361, Oct. 2023.

[6] N. Pahlevan et al., "Seamless retrievals of chlorophyll—A from Sentinel-2 (MSI) and Sentinel-3 (OLCI) in inland and coastal waters: A machine-learning approach," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111604.

[7] K. Shi, Y. Zhang, B. Qin, and B. Zhou, "Remote sensing of cyanobacterial Blooms in inland waters: Present knowledge and future challenges," *Sci. Bull.*, vol. 64, no. 20, pp. 1540–1556, Oct. 2019.

[8] Y. Zhang, R. Ma, H. Duan, S. Loiselle, M. Zhang, and J. Xu, "A novel MODIS algorithm to estimate chlorophyll a concentration in eutrophic turbid lakes," *Ecol. Indicators*, vol. 69, pp. 138–151, Oct. 2016.

[9] M. Werther et al., "A Bayesian approach for remote sensing of chlorophyll—A and associated retrieval uncertainty in oligotrophic and mesotrophic lakes," *Remote Sens. Environ.*, vol. 283, Dec. 2022, Art. no. 113295.

[10] B. P. Page, L. G. Olmanson, and D. R. Mishra, "A harmonized image processing workflow using sentinel-2/MSI and Landsat-8/OLI for mapping water clarity in optically variable lake systems," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111284.

[11] Z. Cao, R. Ma, H. Duan, and K. Xue, "Effects of broad bandwidth on the remote sensing of inland waters: Implications for high spatial resolution satellite data applications," *ISPRS J. Photogramm. Remote Sens.*, vol. 153, pp. 110–122, Jul. 2019.

[12] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 19–34.

[13] K. Fotiadou, G. Tsagkatakis, and P. Tsakalides, "Spectral super resolution of hyperspectral images via coupled dictionary learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2777–2797, May 2019.

[14] X. Zheng, W. Chen, and X. Lu, "Spectral super-resolution of multispectral images using spatial–spectral residual attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404114.

[15] J. Li, Y. Li, L. He, J. Chen, and A. Plaza, "Spatio-temporal fusion for remote sensing data: An overview and new benchmark," *Sci. China Inf. Sci.*, vol. 63, no. 4, Apr. 2020, Art. no. 140301.

[16] M. Parmar, S. Lansel, and B. A. Wandell, "Spatio-spectral reconstruction of the multispectral datacube using sparse recovery," in *Proc. 15th IEEE Int. Conf. Image Process.*, Dec. 2008, pp. 473–476.

[17] Z. Shi, C. Chen, Z. Xiong, D. Liu, and F. Wu, "HSCNN+: Advanced CNN-based hyperspectral recovery from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 939–947.

[18] Y. Zhang, D. Wei, C. Qin, H. Wang, H. Pfister, and Y. Fu, "Context reasoning attention network for image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4278–4287.

[19] C. Wang, J. Jiang, Z. Zhong, and X. Liu, "Spatial-frequency mutual learning for face super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22356–22366.

[20] Y. Wu, R. Dian, and S. Li, "Multistage spatial–spectral fusion network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 12736–12746, Jul. 2025.

[21] C. Zhou, Z. He, L. Zou, Y. Li, and A. Plaza, "HSACT: A hierarchical semantic-aware CNN-transformer for remote sensing image spectral super-resolution," *Neurocomputing*, vol. 636, Jul. 2025, Art. no. 129990.

[22] J. Li, C. Wu, R. Song, Y. Li, and F. Liu, "Adaptive weighted attention network with camera spectral sensitivity prior for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 1894–1903.

[23] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 744–754.

[24] C. Wu, J. Li, R. Song, Y. Li, and Q. Du, "HPRN: Holistic prior-embedded relation network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 11409–11423, Aug. 2024.

[25] C. Shi and T. Nakajima, "Simultaneous determination of aerosol optical thickness and water-leaving radiance from multispectral measurements in coastal waters," *Atmos. Chem. Phys.*, vol. 18, no. 6, pp. 3865–3884, Mar. 2018.

[26] V. Sagan et al., "Monitoring inland water quality using remote sensing: Potential and limitations of spectral indices, bio-optical simulations, machine learning, and cloud computing," *Earth-Science Rev.*, vol. 205, Jun. 2020, Art. no. 103187.

[27] Q. Guan, L. Feng, X. Hou, G. Schurgers, Y. Zheng, and J. Tang, "Eutrophication changes in fifty large lakes on the Yangtze plain of China derived from MERIS and OLCI observations," *Remote Sens. Environ.*, vol. 246, Sep. 2020, Art. no. 111890.

[28] J.-F. Pekel, A. Cottam, N. Gorelick, and A. S. Belward, "High-resolution mapping of global surface water and its long-term changes," *Nature*, vol. 540, no. 7633, pp. 418–422, Dec. 2016.

[29] C. D. Mobley, "Estimation of the remote-sensing reflectance from above-surface measurements," *Appl. Opt.*, vol. 38, no. 36, pp. 7442–7455, 1999.

[30] J.-W. Tang, G.-L. Tian, X.-Y. Wang, X.-M. Wang, and Q.-J. Song, "The methods of water spectra measurement and analysis I: Above-water method," *J. Remote Sensing*, vol. 8, no. 1, pp. 37–44, 2004.

[31] D. Jiang, B. Matsushita, and W. Yang, "A simple and effective method for removing residual reflected skylight in above-water remote sensing reflectance measurements," *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 16–27, Jul. 2020.

[32] Y. Zhang, Y. Zhou, K. Shi, B. Qin, X. Yao, and Y. Zhang, "Optical properties and composition changes in chromophoric dissolved organic matter along trophic gradients: Implications for monitoring and assessing lake eutrophication," *Water Res.*, vol. 131, pp. 255–263, Mar. 2018.

[33] Q. Vanhellemont, "Adaptation of the dark spectrum fitting atmospheric correction for aquatic applications of the Landsat and Sentinel-2 archives," *Remote Sens. Environ.*, vol. 225, pp. 175–192, May 2019.

[34] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[35] L. Chi, B. Jiang, and Y. Mu, "Fast Fourier convolution," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 4479–4488.

[36] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.

[37] K. Zheng et al., "Coupled convolutional neural network with adaptive response function learning for unsupervised hyperspectral super resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2487–2502, Mar. 2021.


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[38] Z. Zhu, J. Hou, J. Chen, H. Zeng, and J. Zhou, "Hyperspectral image super-resolution via deep progressive zero-centric residual learning," *IEEE Trans. Image Process.*, vol. 30, pp. 1423–1438, 2021.

[39] A. P. Trishchenko, J. Cihlar, and Z. Li, "Effects of spectral response function on surface reflectance and NDVI measured with moderate resolution satellite sensors," *Remote Sens. Environ.*, vol. 81, no. 1, pp. 1–18, Jul. 2002.

[40] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[41] J. Xu et al., "Estimation of particulate backscattering coefficient in turbid inland water using sentinel 3A-OLCI image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8577–8593, 2021.

[42] K. Xue, R. Ma, H. Duan, M. Shen, E. Boss, and Z. Cao, "Inversion of inherent optical properties in optically complex waters using Sentinel-3A/OLCI images: A case study using China's three largest freshwater lakes," *Remote Sens. Environ.*, vol. 225, pp. 328–346, May 2019.

[43] W. Chen, X. Zheng, and X. Lu, "Semisupervised spectral degradation constrained network for spectral super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[44] J. Li, Y. Leng, R. Song, W. Liu, Y. Li, and Q. Du, "MFormer: Taming masked transformer for unsupervised spectral reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508412.

[45] L. Tan, R. Dian, S. Li, and J. Liu, "Frequency-spatial domain feature fusion for spectral super-resolution," *IEEE Trans. Comput. Imag.*, vol. 10, pp. 589–599, 2024.

[46] L. Yue, L. Zhang, R. Peng, C. Zeng, H. Duan, and H. Shen, "Understanding the potential, uncertainties, and limitations of spatiotemporal fusion for monitoring chlorophyll a concentration in inland eutrophic lakes," *J. Remote Sens.*, vol. 4, p. 0209, Jan. 2024.

[47] J. F. R. Gower, "Observations of in situ fluorescence of chlorophyll–A in saanich inlet," *Boundary-Layer Meteorol.*, vol. 18, no. 3, pp. 235–245, May 1980.

[48] J. W. Rouse, R. H. Haas, D. Deering, J. Schell, and J. C. Harlan, "Monitoring the vernal advancement and retrogradation (green wave effect) of natural vegetation," NASA, Remote Sens. Center, Texas A&M Univ., College Station, TX, USA, Tech. Rep. RSC 1978-4, 1974.

[49] C. Hu, Z. Lee, R. Ma, K. Yu, D. Li, and S. Shang, "Moderate Resolution Imaging Spectroradiometer (MODIS) observations of cyanobacteria Blooms in Taihu lake, China," *J. Geophys. Res., Oceans*, vol. 115, no. 4, Apr. 2010, Art. no. C04002.

[50] Q. Chen, M. Huang, and X. Tang, "Eutrophication assessment of seasonal urban lakes in China Yangtze river basin using Landsat 8-derived forel-ule index: A six-year (2013–2018) observation," *Sci. Total Environ.*, vol. 745, Nov. 2020, Art. no. 135392.

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